Torsional analysis – Level 2 – Undamped torsional natural frequency analysis

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1. Purpose.

This document is related to the dynamic analysis requirements for rotordynamic pump, especially to torsional analysis.

The procedure to define what level of detail for the dynamic analysis is required to be in accordance with good practice of pump design is described in *G.10.00.09* and *ANSI/HI 9.6.8-2014*: if calculated RUN is higher than 20 and less equal to 50, a **Level 2 – Intermediate method – Torsional rotordynamic analysis – Undamped torsional natural frequency analysis** is required**.**

This document describes how to perform an undamped natural frequency (torsional) analysis and a forced response (torsional) analysis.

2. Scope.

This procedure is applicable to both VS and OH or BB pumps.

3. Definitions.

BB	between bearings (API610)
C_L, C_S, C_G, C_T, C_R	factors for fatigue analysis
D, d	diameter
DOF	degree of freedom
E	elastic modulus
F, f	force or frequency
FEA	finite element analysis
G	shear modulus
GUI	graphical user interface
IDmb	inner bending mass diameter
IDsb	inner bending stiffness diameter
K_f	fatigue stress concentration factor

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K_t	torsional stiffness or geometric stress concentration factor (in fatigue analysis)
J	moment of inertia
L	length
M, m	mass
N	rotating speed
ODmb	outer bending mass diameter (Madyn 2000)
ODsb	outer bending stiffness diameter (Madyn 2000)
OH	overhung (API610)
P	power
PF	penetration factor (API684)
R	risk number (ANSI HI 9.6.8-2014)
ROTx	rotational degree of freedom around x axis in cartesian coordinate system
ROTy	rotational degree of freedom around y axis in cartesian coordinate system
ROTz	rotational degree of freedom around z axis in cartesian coordinate system
RUN	R multiplied by U (ANSI HI 9.6.8-2014)
S_n', S_n	endurance limit and fatigue strength
S_y	yield strength
S_u	ultimate strength
T	torque
U	uncertainty value (ANSI HI 9.6.8-2014)
VFD	variable frequency driver
VS	vertically suspended (API610)
v	Poisson's coefficient
ρ	density
σ	normal stress
τ	shear stress

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ω excitation speed

Ω rotational speed

Campbell diagram it indicates the natural frequencies within the excitation natural frequency range and displays the margin between potential torsional resonances identified the analysis versus those excitation frequencies (1x running speed, 2x running speed, vane pass, 1x and 2x electrical line frequency, etc…).

Critical Speed shaft rotational speed at which the rotor-bearing-support system is in a state of resonance.

Goodman diagram it predicts the fatigue life at a given stress ratio (alternating and mean stresses), based on the material properties (ultimate strength and endurance strength).

Station boundary points of a discrete element.

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4. General procedure: description.

A torsional analysis must be performed on the complete shaft train (pump, driver, coupling, gears, etc.). Torsional vibration is fluctuating angular motion that causes twisting in couplings and shafting system of rotating machinery. The torsional analysis (1) simulates how the pump, motor, coupling, and gear (if included) operate dynamically in twist excitation when run together as a system, and (2) identifies torsional natural frequencies.

Identified problems or resonances are resolved by changing the mass or inertia of components in the system (impellers, couplings, flywheels) or by changing the torsional stiffness of low-stiffness components, for example, by changing shaft diameters or changing the coupling type in the specification.

High torsional vibration may not be detected by typical field vibration analysis methods. High-order field vibration analysis methods, such as specially instrumented shaft contact devices or strain gauges attached to radio telemetry, are required to troubleshoot a torsional problem. Excessive or unusual coupling or bearing wear, and/or high levels of gear noise are indicators of a torsional problem. Excessive torsional vibration can lead, or contribute, to the following types of problems:

- Damaged couplings or other coupling problems
- Gear wear, noise or tooth failure
- Shaft fatigue or failure
- Damage at shaft attachments, such as keys or fits
- Loosening of motor windings

The assumption is that the analysis is performed at the design point at steady state operation (flow, pressure, speed, temperature, etc…).

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5. Methodology, analysis and interpretation of results.

Methodology.

Software/Tools:

- general purpose FEA programs or specialty rotordynamic programs;

The tool described in this procedure is:

- Madyn 2000

Model.

The main assumption in the finite element analysis (FEA) is that a continuous structure can be approximated by describing it as an assembly of discrete elements, each with a number of boundary points that are referred to as nodes (or stations). Elements have one nodal degree of freedom (DOF): rotation around shaft axis (ROT_z).

The complete train, typically motor-coupling-pump, must be modelled because it is torsionally coupled due to the substantial torsional stiffness of the coupling.

Pump rotor model: shaft accounts for stiffness, mass and inertia, while impellers, sleeves, etc…, account for mass and inertia only. An exception is for impeller, hub, or sleeve fit-ups with a press-fit and or a key that extends in axial length more than two thirds the diameter of the shaft at that location. In such circumstances the shaft torsional stiffness should be considered unaffected over a length on each end equal to one third the length of the fit-up. The middle one third should have its torsional stiffness assessed based in the average outside diameter (within the middle one-third zone) of the fitted component.

A portion of the fluid inside the impeller passage contributes to the overall polar mass moment of inertia. Typically, the torsionally “active” fluid contributes less than 5% added moment of inertia to the impeller, and in many cases less than 1%. Therefore, it could be neglected.

Motor rotor mass elastic data (stiffness and polar moment of inertia) needs to be obtained from the motor vendor.
Motor core is considered as distributed mass over its length. Stiffness contribution of the motor core is considered using equivalent inner and outer diameters.

Coupling mass elastic data (stiffness and polar mass moment of inertia) needs to be obtained from the coupling vendor.
When a shaft is joined to a nonrigid coupling or impeller by an interference fit, the shaft should be assumed to twist freely over a length equal to one-third of the overlap. When a shaft is joined to a nonrigid coupling or impeller by a keyed joint, the shaft should be assumed to twist freely over a length equal to two-third of the overlap.

A nearly unanimous opinion is that hydraulic couplings should be treated as zero spring rate elements that effectively divide the assembly into two independent torsional systems. Another opinion is that the zero-spring rate is an approximation since, in reality, restoring forces in the fluid yield a small but finite spring rate. These restoring forces are generated by centrifugal forces in the fluid and are, therefore, proportional to the square of running speed.
Thus, when modelling assemblies containing hydraulic couplings, the user has two options. Firstly, the machine can be treated as a single entity and the equations or data from vendor can be utilized to determine the coupling's effective stiffness. On the other hand, the most common opinion is that the coupling can be assumed to have zero stiffness and the two resultant systems can be modelled separately.

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For torsional analyses involving speed-changing devices (gear drives, belt drives, etc.) the rotational inertia and stiffness properties should be modified (according to the square of the speed ratio) to address the effects of speed modifying device.
The parameter values for elements on the low speed shaft are unchanged. However, elements in the high speed shaft must be transformed via the following equations:

$$J_{eq} = J \cdot N^2$$
$$k_{eq} = k \cdot N^2$$

Where:

- $J_{eq} = Equivalent\ inertia\ referenced\ to\ low\ speed\ shaft$
- $J = Actual\ inertia$
- $N = Gear\ ratio\ (N > 1.0)$
- $k_{eq} = Equivalent\ stiffness\ referenced\ to\ low\ speed\ shaft$
- $k = Actual\ stiffness$

Although gear teeth have inherent flexibility, for most practical cases, they can be considered to be torsionally rigid.

Analysis.

Due to low torsional damping present in motor-coupling-pump shaft trains, an undamped natural frequencies analysis can be performed.
Torsional natural frequencies are independent of operating speed (typically no component has speed-dependent torsional stiffness). However, certain types of flexible coupling have torque-dependent stiffness that should be accounted for, e.g., by setting the coupling stiffness based on the torque present at the pump operating speed.

Forced response analysis determines the rotor's harmonic response to applicable steady state excitations (e.g., dynamic torque functions). The assumptions made in the analysis regarding the magnitude of excitation and the degree of damping shall be clearly stated.
Appropriate fatigue properties and stress concentrations shall be used.
Mode shapes, angular displacements, and stresses are calculated at critical points along the drive train. All forcing functions should be applied using the worst-case scenario of location and direction.

Interpretation of results.

Results of an undamped natural frequency are best presented in form of a Campbell diagram (Figure 1).

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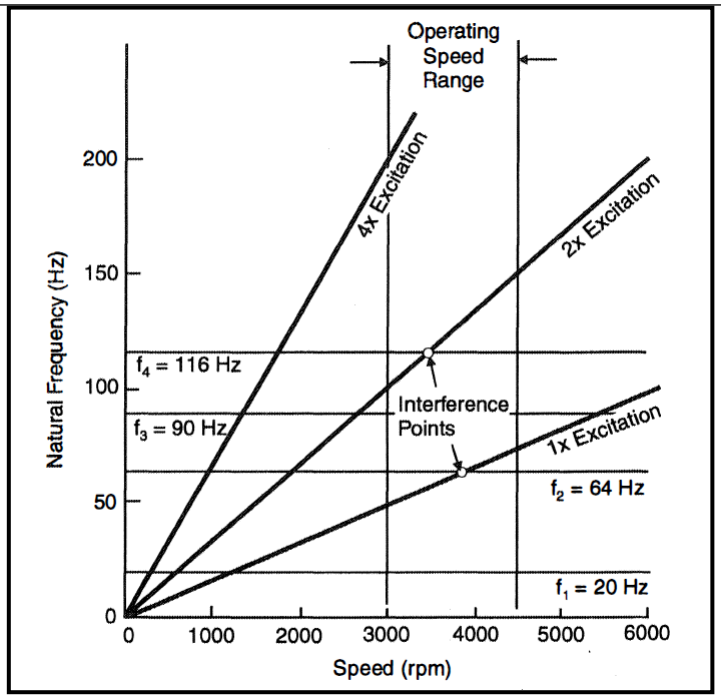

Figure 1. Representative Campbell Diagram.

Natural frequencies are plotted as horizontal lines. Operating speed range is designated by vertical lines. The upward sloping lines are harmonics of speed that represent the system's potential excitations. If excitation sources are not a function of speed, such as line frequency, they are represented as straight lines. Intersections between these lines and the natural frequency lines that occur within the operating speed range are referred to as interference points and are indicators of potential resonances.

6. Acceptance criteria.

As indicated in API 610 12th Edition §6.9.2.6, the torsional natural frequencies of the complete train shall be at least 10% above or 10% below any possible excitation frequency within the operating speed. In case the desired separation margin is not fulfilled, the natural frequency of concern may, for example, be shift by means of modifying the torsional stiffness of the coupling. If an acceptable margin cannot be obtained, then a steady state damped response analysis shall be performed to demonstrate that resonances have no adverse effect on the complete train (§6.9.2.7). The forced response analysis should generate dynamic torsional torques and stresses to evaluate the mechanical integrity of the components (shafts and coupling). The maximum torques shall be used to evaluate the peak torque capability of coupling components, gearings and interference fits of components, such as coupling hubs, and shaft locations with stress raisers (§6.9.2.11). Figure 2 summarizes the torsional analysis procedure.

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For synchronous motors all natural frequencies less than 2X line frequency will be excited by the pulsating torque during startup. Then, there will be no resonant speeds if all system natural frequencies are above 2X line frequency. If not, a transient forced response analysis will be required.

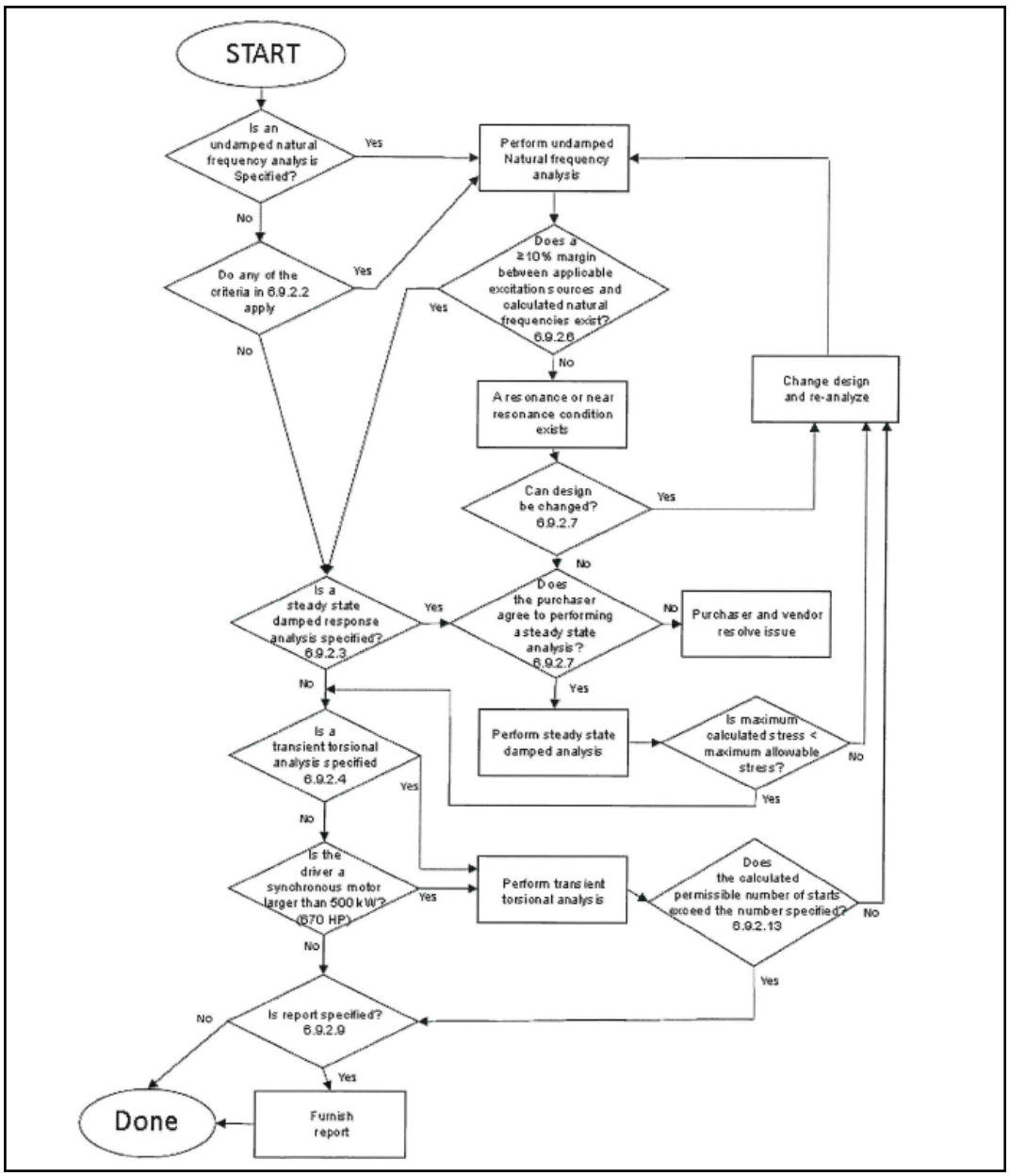

Figure 2. Torsional Analysis Flow Chart.

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G.10.00.09 - Guidelines for dynamic analysis complexity level

Mod.401 – Spreadsheet linked to procedure G.10.00.09 (in agreement with ANSI HI 9.6.8)

Mod.407 - Torsional analysis Campbell diagram

Mod.408 - Torsional analysis shaft fatigue verification

Mod.409 - Torsional analysis coupling verification

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Annex A. Input data requirements

The following information is typically required for a torsional rotordynamic analysis:

1. Pump shaft dimensions (drawing preferred)
2. Rotor material of construction
   - Shaft
   - Impellers
   - Sleeved
   - Coupling
3. Mass and inertia (polar) for all rotating components
   - Impellers
   - Sleeves
   - Mechanical seal rotating components

Note: mass and inertia can be estimated from component geometry if necessary.

4. Assembly information
   - Axial location of all rotating components on the shaft
   - Diametral fit of all rotating components on the shaft (loose or press, and if press, how much diametral interference)
   - Length and location of all keys
5. Coupling information (drawing preferred, and usually available from vendor)
   - Manufacturer, type and size
   - Coupling torsional stiffness (as a function of speed if applicable)
   - Details of spacer (length, mass and inertia)
   - Coupling hub fit-up length and degree of press fit, length of key
6. Motor information
   - Rotor drawing
   - Rotor mass and inertia (polar)
   - Rotor core fit-up on shaft (loose or press, and if press, how much diametral interference)
   - Motor fan mass and inertia (polar)
   - Motor fan fit-up on shaft
   - Length and location of key
   - Torsional stiffness of the motor stub shaft between the motor core end and coupling hub

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7. Diesel information

- Polar inertia of all rotating and reciprocating components (i.e., flywheel, coupling, pistons, etc.)
- Torsional stiffness of the shaft
- Number of power strokes per revolution
- Fit of coupling to shaft (clearance or press fit)
- Running speed

8. Turbine information

- Rotor drawing or sketch showing key shaft dimensions and envelope dimensions of each turbine wheel
- Rotor materials of construction – shaft, turbine wheels, sleeves
- Mass and inertia (polar) for rotation components
- Number of blades on turbine wheel

9. Right-angle gear information

- Gear drawings
- Shaft torsional stiffnesses
- Gear polar inertias
- Gear ratio
- Number of gear teeth

10. Assembly information

- Axial location of all rotating components on the shaft
- Diametral fit of all rotating components on the shaft (loose or press)
- Length and location of all keys

General information needed for torsional analysis:

1. Excitation information

- Pump rotational speed or operating speed range
- Number of vanes on impeller (both first stage and intermediate stages)
- Number of vanes on diffuser

2. Required separation margins and/or specifications to be met

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Annex B. Excitation sources.

Excitation of torsional natural frequencies can come from many sources, which might or might not be a function of running speed and should be considered in the analysis.

- The most ***common excitations*** are at once per revolution (1x) and twice per revolution (2x). These are generic sources that can arise via various mechanisms. 1x excitations can be generated by conditions such as rotating unbalance, eccentricity, and misalignment. Excitations of 2x are usually due to misalignment, ellipticity, or certain non-circular shaft cross-sections such as keyways.
- ***Gears*** generate pulsations at several different frequencies. Errors in the generation of the gear teeth and in the mounting of the gear hub on the shaft can lead to unbalance, eccentricity and/or misalignment. Such errors lead at frequency once per revolution of the shaft (1x). Any errors that result in gear ellipticity generate torque variations at twice shaft speed (2x).

  Gears can produce disturbances at their meshing frequency and higher harmonics of it. For each shaft, meshing frequency is equal to the number of teeth on that shaft's gear multiplied by shaft rpm.

  Whenever gears are present in a system the interference diagram for each shaft should include three excitation lines: 1x, 2x and n_{teeth}. Higher harmonics of gear mesh frequency can be dismissed because of their negligible magnitudes.
- Another common excitation source is the ***vaned impeller***. In pump impellers, torque variations occur at blade-pass frequency due to pressure disturbances resulting from vanes passing a stationary object (n_{blades}).

  Impellers operate within casings that contain several equally spaced obstacles that can generate pressure fluctuations when passed by an impeller blade. It occurs in configurations employing vaned diffusers or volutes with multiples cutwaters (n_{vanes}).

  Disturbances can also be generated each time a rotor blade passes by a stationary vane or cutwater:

$$n = {}^{n_{blade}} \cdot {}^{n_{vanes}}/_{c_h}$$

  - $n = excitation\ order\ number$
  - $c_h = highest\ common\ factor\ of\ n_{blade}\ and\ n_{vanes}$

  A vaned impeller operating within a vaned diffuser can generate three separate excitations.
- The pulsating torques that arise in ***electric motors*** can be separate in two categories; those that occur when the machine is running at constant speed and those that transpire

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when the machine is accelerating or exposed to an electrical fault (not in the aim of a level 2 torsional analysis).

Motors have an infinite number of designs and each one has its own unique excitation mechanisms. The best source of information regarding excitations produced by electric machines is the manufacturer.

At steady speed, most AC motors produce fluctuations at line frequency (1xLF) and twice line frequency (2xLF). Thus, systems containing AC machines have horizontal excitation lines in the Campbell diagram. Many machines, also, create oscillations having an order number equal to the number of magnetic poles (n_{poles}).

In general, the excitations for systems utilizing AC machines are: 1xLF, 2xLF and n_{poles} times the running frequency.

DC machines generate excitations that are relatively small during steady operation. Therefore, DC motors can be neglected in the torsional vibration calculation.

- Variable speed motors are controlled by ***variable frequency drives (VFD)*** which alter motor speed by varying the electrical frequency supplied to the motor's terminals. These drives add torsional excitations.

  There are different types of VFD: static frequency converter which electronically transforms the constant line frequency to the desired electrical frequency for driving the motor; Kramer drives that control motor speed by varying the motor's slip; and impressed voltage inverter with PWM (Pulse Width Modulation) strategy.

  Most *static converters* consist of a rectifier in series with an inverter. The rectifier converts the AC signal at line frequency into a DC signal. The DC is subsequently converted back into an AC signal at the desired frequency by the inverter. The outputs of inverter are not perfect sinusoids, periodic variations in the driven motor's output torque are created.

  The speed of induction motors is usually varied via control of the stator frequency. In these drives the excitation frequencies are given by:

$$f_{excitation} = n \cdot k \cdot f_{ele}$$

  - $f_{ele} = electrical\ output\ by\ inverter$
  - $n = number\ of\ pulses$
  - $k = 1, 2, 3 \ldots$

  In both synchronous and induction motors, the motor speed is approximately related to the supplied electrical frequency by:

$$f_{motor} = {f_{ele}}/{\left(n_{poles}/2\right)}$$

  The excitations are:

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$$f_{excitation} = n \cdot k \cdot \left(n_{poles}/2\right) \cdot f_{motor}$$

The value of n is either 6 or 12 (for VSI drivers n = 1, 2 and k = 1).

Kramer drives produce pulsations at harmonics of the slip frequency, that is the difference between the line frequency and the motor speed. The generated harmonics are multiples of 6 or 12, dependent on the number of pulses in the drive. The pulsations merely occur at line and twice line frequency since the electrical supply frequency is not variable.

In addition to the pulsations generated by the variable frequency drive, the normal fluctuations produced by AC machines are also present.

Impressed voltage converter, due to the type of converter and modulation strategy, do not present harmonics with frequency multiple of the motor operating frequency (5th, 7th, 11th, 13th, …). The torque harmonics spectrum consists of harmonic torques centered on inverter switching frequency and its multiples with side bands dependent by motor operating frequency.

The parameters that have an influence on harmonic torques are mainly:

- inverter switching frequency
- motor electrical parameters
- motor operating frequency

Depending on the characteristics of the inverter, at any operating point, none of the torque pulsations harmonics exceed 1% of the motor rated torque (RMS value, Figure 3).

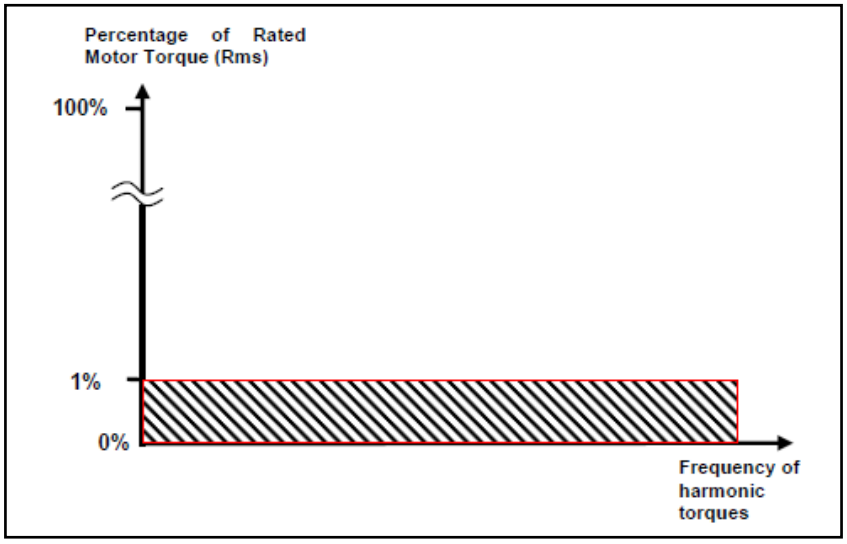

Figure 3. Spectrum of toque pulsations harmonics.

- ***AC synchronous motors*** generate torque pulsations during the starting process. Unlike induction motors, synchronous motors are not self-starting. They are normally equipped with squirrel-cage windings which provide starting torque and damping during steady state

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running. During starting procedure pulsating torques are created due to the fact that synchronous motor rotors contain salient poles that are magnetic enclosed by field coils. The resulting asymmetry causes the motor's output to vary as a function of rotor position.

The frequency of the torque pulsation is the frequency at which the stator's rotating magnetic field passes a rotor pole. Since the stator's magnetic field rotates at synchronous speed, the excitation frequency is a function of the difference between synchronous speed and rotor speed, which is known as slip speed.

$$f_{slip} = f_{line} \cdot (N_s - N)/N$$

Where:

$f_{slip} = slip\ freqeuncy\ (Hz)$

$f_{line} = line\ frequency\ (Hz)$

$N_s = 2 \cdot 60 \cdot \frac{f_{line}}{N_{poles}} = synchronous\ speed\ (rpm)$

$N = rotating\ speed\ (rpm)$

$N_{poles} = number\ of\ poles$

It is seen that the frequency of torque. pulsations, twice slip frequency, decreases as rotor speed increases. Thus, at zero speed, the excitation frequency is equal to two times line frequency. As the motor is accelerated, the excitation frequency decreases linearly until it reaches zero when the rotor achieves synchronous speed (Figure 4).

Most systems driven by synchronous motors will experience resonances with all natural frequencies below twice the line frequency during starting. This is a significant source of potential problems since most practical turbomachinery drive trains have several natural frequencies within this range.

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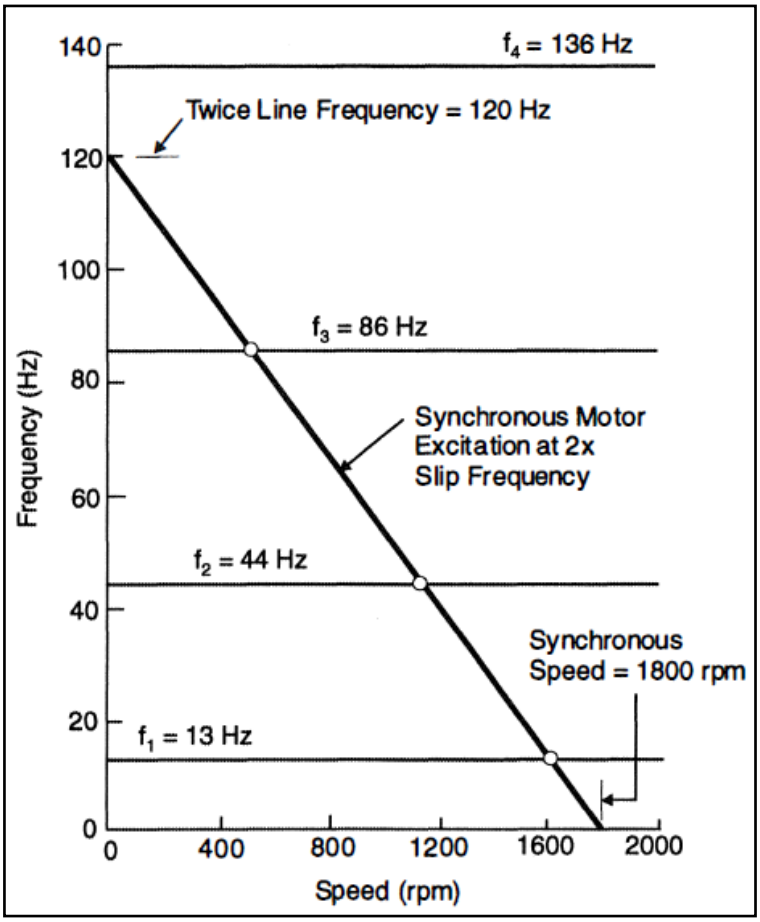

Figure 4. Synchronous Motor Campbell Diagram.

- ***2-cycle engines***: n times the rotor speed where n is the number of power strokes per revolution;
- ***4-cycle engines***: n and 0.5 times the rotor speed where n is the number of power strokes per revolution.

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Table 1. Summary excitation Sources and Frequencies

Excitation source	Excitation Frequencies
Generic 1x (unbalance, eccentricity, misalignment, etc.)	- 1 x Speed
Generic 2x (misalignment, ellipticity, etc.)	- 2 x Speed
Gear Mesh consisting of Pinion with N_P teeth mating with Gear having N_G teeth	Pinion shaft - One x Pinion Speed - Two x Pinion Speed - N_P x Pinion Speed Gear shaft - One x Gear Speed - Two x Gear Speed - N_G x Gear Speed
Impeller with N_R blades rotating inside Casing with N_S cutwaters	- N_R x Speed - N_S x Speed - N_R x N_S / c_h x Speed
AC Motor With N_P Poles (Fixed Frequency or Kramer Drive)	- Line Frequency - Twice Line Frequency - N_P x Speed
AC Motor With N_P Poles (Variable Frequency Drive Controlling Stator)	- ½ N_P x Speed - N_P x Speed
Variable Frequency Drive (Stator Frequency control with N Pulses driving AC Motor with N_P poles	- ½ N x N_P x Speed - N x N_P x Speed - 1.5 N x N_P x Speed - 2 N x N_P x Speed
Static Kramer Drive with N Pulses	- N x Slip Frequency - 2 x N x Slip Frequency
Synchronous Motor	- 2 x Slip Frequency
2-cycle engine With N_{PS} power strokes per revolution	- N_{PS} x Speed
4-cycle engine With N_{PS} power strokes per revolution	- ½ x Speed - N_{PS} x Speed

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Annex C. Torsional rotordynamic analysis with MADYN 2000.

2.1. Madyn 2000, System object

In MADYN 2000 the model, the loads and further analysis parameters as well as analysis results are stored in the system object. The model can consist of several objects such as shafts, couplings, bearings.

A vibration analysis consists of the following steps:

- Modelling a system,
- defining loads on the system,
- running the analyses,
- plotting, listing and interpretation of results.

Each of these steps can be controlled from the system explorer.

The modelling process can be started from the “File” menu of the main MADYN 2000 window by the menu item “Create new System” (Figure 5).

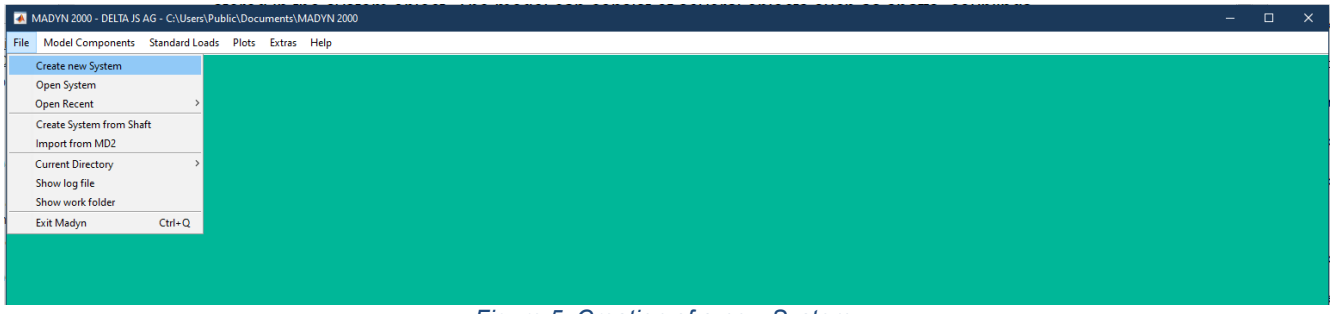

Figure 5. Creation of a new System.

The “Create new System” item leads to the opening of the *system explorer* and the *system GUI* (Figure 6). The *system GUI* has buttons to add shafts, gears and flexible couplings. The connection button opens a GUI to connect shafts and couplings. The pull-down menu “TLA case” allows to select the type of analysis (torsion, lateral…).

The “add …” buttons in the *system GUI* open the “Cancel”/ “Create New”/ “File” window to either create a new object via its GUI or to select a file with the corresponding object (shaft, gear or flexible coupling) to be loaded.

Usually, shaft objects are uploaded from a file.txt following the procedure in *2.2 Madyn 2000, Shaft object* and connection objects are created as new following instructions in *2.4 Madyn 2000, Connection objects*.

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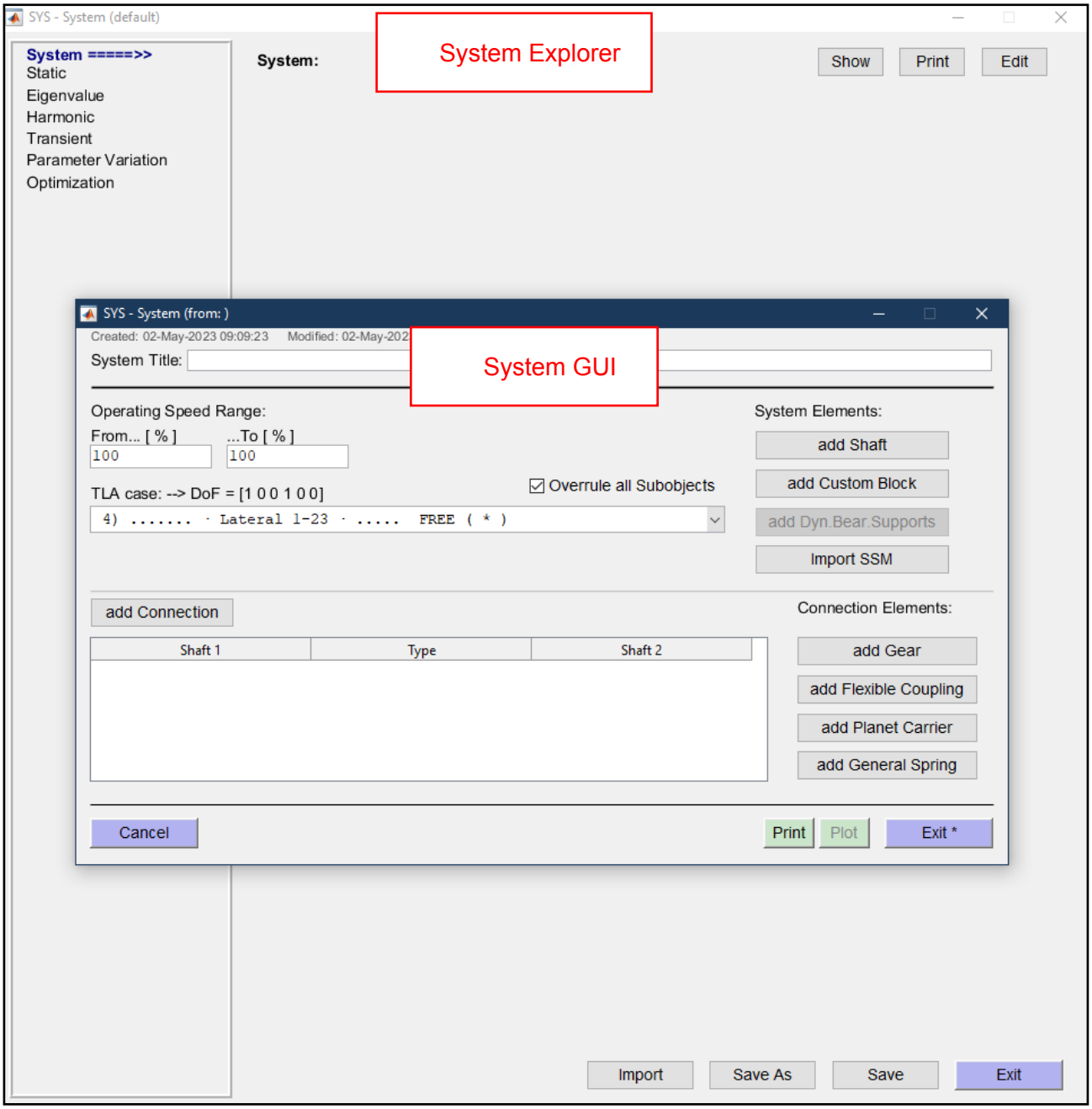

Figure 6. System explorer and system GUI.

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Figure 7 is an example of a *system GUI* for torsional analysis:

- The operating speed range goes form 90% (or 85%) to 110% (or 115%) as recommended in API 610 6.9.2.6.
- TLA case is set on torsional analysis.
- Shafts constituting the complete train are uploaded.
- Connection elements (gears and flexible coupling) are created.
- Connections between shafts are defined.

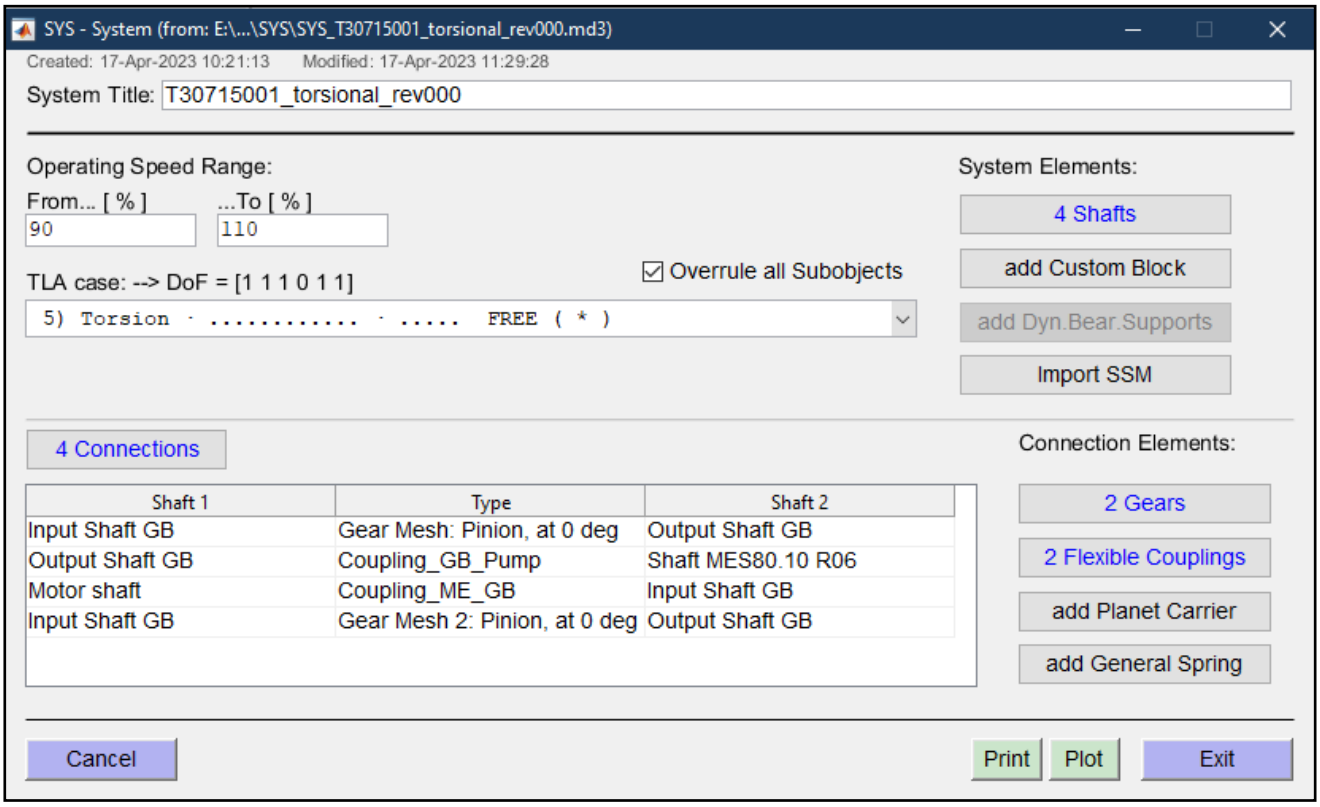

Figure 7. Example of System GUI.

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2.2. Madyn 2000, Shaft object

The creation of a rotor model starts from a spreadsheet that can be found in the directory *C:\Users\Public\Documents\MADYN 2000\Examples\Shaft_Input_w_EXCEL.*

The spreadsheet has different columns that need to be compiled.

- Columns “Length” and “ODsb”, that stands for outer bending stiffness diameter, are the mainly used. Each element should have a ratio L/D less than 0.5 (for example, length defined in Figure 8 is not correct).
- If there is a hollow section, the internal diameter should be defined in column “IDsb”, that stands for inner bending stiffness diameter.
- If there is a change in diameter in an element, the value in the “Conical” column needs to be changed to 1. 0 stands for a cylindrical element, while 1 for a conical one. To define a conical section, it is needed of two rows. In the first one the left-side diameter will be put, while in the other one the right-side diameter.
- If there are parts in the rotor that gives mass contribution but do not add stiffness, the values in column “ODmb” must be changed. For default, these values are equal to “ODsb”. The same argument counts for the inner diameter: the values in column “IDmb” must be changed.
- Stations should be defined where there is the center of gravity of mounted parts on shaft, such as impellers and couplings. Lumped masses and moment of inertia can be inserted through columns “M”, “Thetap” and “Thetae”. Column “Centre” defines the offset of the lumped mass from the station assigned.
- Columns “addmass”, “addinertiap” and “addinertiab” defines mass and moments of inertia per unit length. “addmass” can be used to define the fluid swept by the rotor during operation.
- Column “Denotation” is useful to highlight station of relevance. It will be helpful later in the model to locate these stations.

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Shaft Model Data for MADYN 2000 14/04/2023

Disclaimer:
No responsibility for the correctness of the calculations in this spreadsheet is taken by DELTA JS AG.

Shaft Summary:		
Density:	7'850	[kg/m3]
Length:	2'720.5	[mm]
Mass:	2'687.3	[kg]
Thetap:	201.28	[kgm2]

Auxiliary values needed for the calculation of total shaft mass and polar moment of inertia:

Shaft Title

* #	Conical	Length	IDsb	IDmb	IDst	ODsb	ODmb	ODst	M	Thetap	Thetae	Centre	addmass	addinertiap	addinertiab	Temperature	Denotation	Sec. M	SEC Thetap	Add M	Add Thetap
*	[-]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[kg]	[kgm2]	[kgm2]	[mm]	[kg/m]	[kgm2/m]	[kgm2/m]	[°C]	[-]	[kg]	[kgm2]	[kg]	[kgm2]
1	0	166.7	0.0	0.0	0.0	130.0	130.0	130.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		17.37	0.037	0.00	0.00
2	0	23.3	0.0	0.0	0.0	130.0	130.0	130.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		2.43	0.005	0.00	0.00
3	0	60.0	0.0	0.0	0.0	130.0	130.0	130.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40	'Coupling connection'	6.25	0.013	0.00	0.00
4	0	200.5	0.0	0.0	0.0	140.0	140.0	140.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		24.23	0.059	0.00	0.00
5	0	15.0	0.0	0.0	0.0	156.0	156.0	156.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		2.25	0.007	0.00	0.00
6	0	45.0	0.0	0.0	0.0	172.5	172.5	172.5	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		8.26	0.031	0.00	0.00
7	0	44.5	0.0	0.0	0.0	172.5	172.5	172.5	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40	'DE bearing'	8.16	0.030	0.00	0.00
8	0	73.0	0.0	0.0	0.0	200.0	200.0	200.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		18.00	0.090	0.00	0.00
9	0	226.4	0.0	0.0	0.0	247.4	247.4	247.4	50.0	3.00	1.50	-100.00	1500.0	150.0	75.0	40	'End windings'	85.43	0.654	339.60	33.96
10	0	205.1	0.0	0.0	0.0	247.4	247.4	247.4	0.0	0.00	0.00	0.00	1500.0	150.0	75.0	40		77.38	0.592	307.58	30.76
11	0	205.1	0.0	0.0	0.0	247.4	247.4	247.4	0.0	0.00	0.00	0.00	1500.0	150.0	75.0	40		77.38	0.592	307.58	30.76
12	0	205.1	0.0	0.0	0.0	247.4	247.4	247.4	0.0	0.00	0.00	0.00	1500.0	150.0	75.0	40	'Centre of lamination'	77.38	0.592	307.58	30.76
13	0	205.1	0.0	0.0	0.0	247.4	247.4	247.4	0.0	0.00	0.00	0.00	1500.0	150.0	75.0	40		77.38	0.592	307.58	30.76
14	0	226.4	0.0	0.0	0.0	247.4	247.4	247.4	0.0	0.00	0.00	0.00	1500.0	150.0	75.0	40		85.43	0.654	339.60	33.96
15	0	147.0	30.0	30.0	30.0	200.0	200.0	200.0	50.0	3.00	1.50	100.00	0.0	0.0	0.0	40	'End windings'	35.44	0.181	0.00	0.00
16	0	50.0	30.0	30.0	30.0	172.5	172.5	172.5	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		8.90	0.034	0.00	0.00
17	0	82.0	30.0	30.0	30.0	149.0	149.0	149.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		10.77	0.031	0.00	0.00
18	0	136.5	30.0	30.0	30.0	149.0	149.0	149.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		17.93	0.052	0.00	0.00
19	0	60.0	30.0	30.0	30.0	149.0	149.0	149.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		7.88	0.023	0.00	0.00
20	0	32.0	30.0	30.0	30.0	149.0	149.0	149.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		4.20	0.012	0.00	0.00
21	0	44.0	30.0	30.0	30.0	131.0	131.0	131.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		4.41	0.010	0.00	0.00
22	0	44.0	30.0	30.0	30.0	131.0	131.0	131.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40	'NDE bearing'	4.41	0.010	0.00	0.00
23	1	140.0	30.0	30.0	30.0	131.0	131.0	131.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		12.82	0.027	0.00	0.00
23	1	0.0	30.0	30.0	30.0	120.0	120.0	120.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		0.00	0.000	0.00	0.00
24	0	84.0	30.0	30.0	30.0	90.0	90.0	90.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	40		3.73	0.004	0.00	0.00
25	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	20		0.00	0.000	0.00	0.00
26	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	20		0.00	0.000	0.00	0.00
27	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	20		0.00	0.000	0.00	0.00
28	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	20		0.00	0.000	0.00	0.00
29	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	20		0.00	0.000	0.00	0.00
30	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.00	0.00	0.0	0.0	0.0	20		0.00	0.000	0.00	0.00

Figure 8. Example of spreadsheet for rotor input data.

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After compiling the spreadsheet, the part including rotor geometry should be copied and saved as a file.txt. It is important to use a proper name for the file because Madyn 2000 recognize the file as a shaft object only if the file name has a prefix SFT_ (i.e. SFT_example).

SFT_shaft.txt - Blocco note di Windows

File Modifica Formato Visualizza ?

```
Shaft Title
* #     Conical Length  IDsb   IDmb   IDst   ODsb   ODmb   ODst   M      Thetap  Thetae  Centre  addmass addinertiap    addinertiab    Temperature    Denotation
*       [-]     [mm]    [mm]   [mm]   [mm]   [mm]   [mm]   [mm]   [kg]   [kgm2]  [kgm2]  [mm]    [kg/m]  [kgm2/m]       [kgm2/m]       [°C]    [-]
1       0       166.7   0.0    0.0    0.0    130.0  130.0  130.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
2       0       23.3    0.0    0.0    0.0    130.0  130.0  130.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
3       0       60.0    0.0    0.0    0.0    130.0  130.0  130.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40      'Coupling connection'
4       0       200.5   0.0    0.0    0.0    140.0  140.0  140.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
5       0       15.0    0.0    0.0    0.0    156.0  156.0  156.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
6       0       45.0    0.0    0.0    0.0    172.5  172.5  172.5  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
7       0       44.5    0.0    0.0    0.0    172.5  172.5  172.5  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40      'DE bearing'
8       0       73.0    0.0    0.0    0.0    200.0  200.0  200.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
9       0       226.4   0.0    0.0    0.0    247.4  247.4  247.4  50.0   3.00    1.50    -100.00 1500.0  150.0   75.0   40      'End windings'
10      0       205.1   0.0    0.0    0.0    247.4  247.4  247.4  0.0    0.00    0.00    0.00    1500.0  150.0   75.0   40
11      0       205.1   0.0    0.0    0.0    247.4  247.4  247.4  0.0    0.00    0.00    0.00    1500.0  150.0   75.0   40
12      0       205.1   0.0    0.0    0.0    247.4  247.4  247.4  0.0    0.00    0.00    0.00    1500.0  150.0   75.0   40      'Centre of lamination'
13      0       205.1   0.0    0.0    0.0    247.4  247.4  247.4  0.0    0.00    0.00    0.00    1500.0  150.0   75.0   40
14      0       226.4   0.0    0.0    0.0    247.4  247.4  247.4  0.0    0.00    0.00    0.00    1500.0  150.0   75.0   40
15      0       147.0   30.0   30.0   30.0   200.0  200.0  200.0  50.0   3.00    1.50    100.00  0.0     0.0     0.0    40      'End windings'
16      0       50.0    30.0   30.0   30.0   172.5  172.5  172.5  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
17      0       82.0    30.0   30.0   30.0   149.0  149.0  149.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
18      0       136.5   30.0   30.0   30.0   149.0  149.0  149.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
19      0       60.0    30.0   30.0   30.0   149.0  149.0  149.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
20      0       32.0    30.0   30.0   30.0   149.0  149.0  149.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
21      0       44.0    30.0   30.0   30.0   131.0  131.0  131.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
22      0       44.0    30.0   30.0   30.0   131.0  131.0  131.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40      'NDE bearing'
23      1       140.0   30.0   30.0   30.0   131.0  131.0  131.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
23      1       0.0     30.0   30.0   30.0   120.0  120.0  120.0  0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
24      0       84.0    30.0   30.0   30.0   90.0   90.0   90.0   0.0    0.00    0.00    0.00    0.0     0.0     0.0    40
```

Figure 9. Example of SFT_file.txt

In order to create the shaft object from the SFT file, it is necessary to follow the path in Figure 10.

If it is the first time that an SFT file will be imported, before importing the file, it is necessary to upload in Madyn the SIF_Shaft.txt file from the command “SFT – Ask for SIF and Load”. The folder containing this file is the same of the excel file:

- *C:\Users\Public\Documents\MADYN 2000\Examples\Shaft_Input_w_EXCEL.*

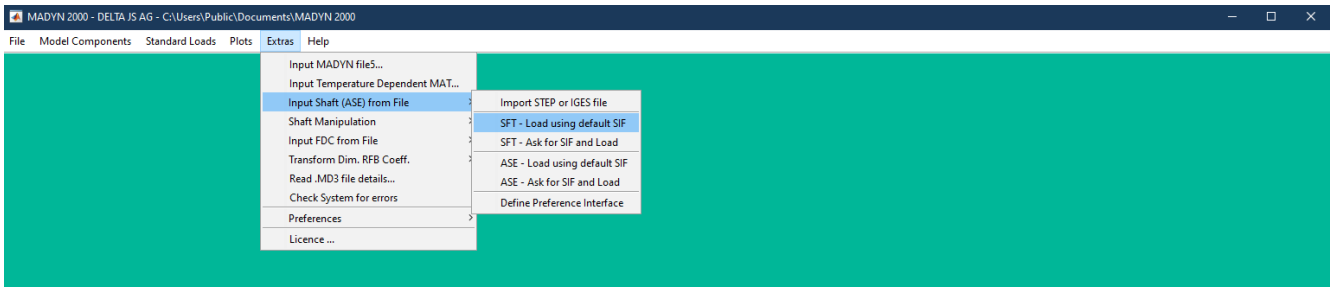

Figure 10. Import path for shaft object.

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Selecting the correct SFT file to open, Madyn 2000 generates the shaft object that can be modified through the interface in Figure 11.

The *shaft GUI* contains default values for the material and TLA case. It is important to define the correct rotating speed of the shaft.

The shaft object can be saved through right click or with the Exit command.

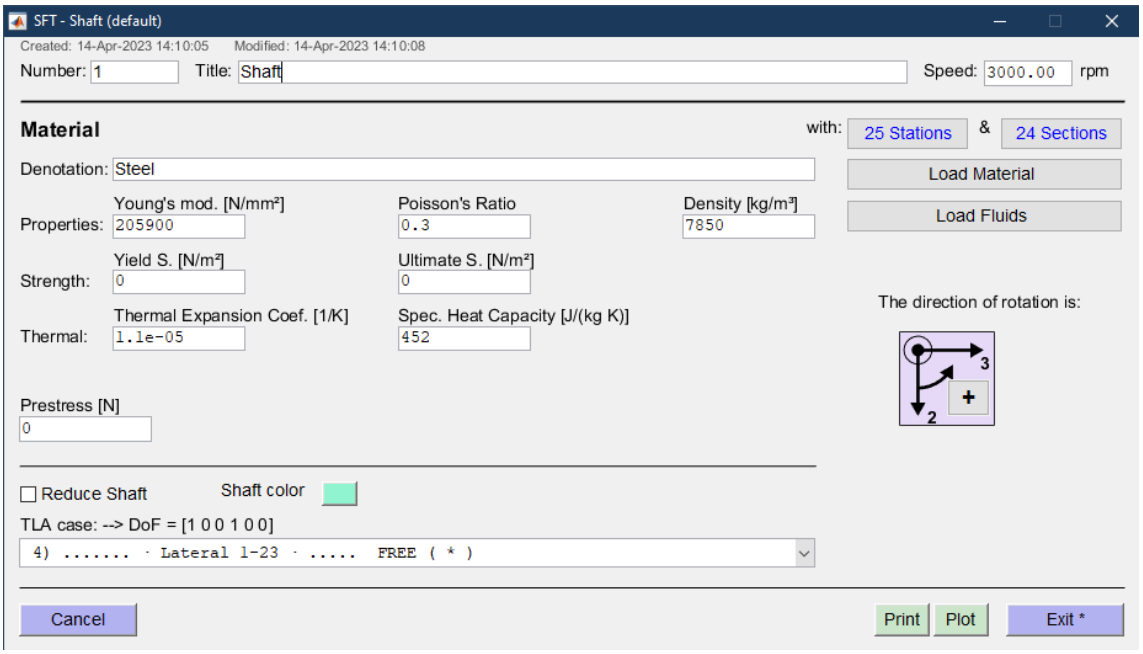

Figure 11. Shaft GUI.

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2.3. Electric motor model

The difference between a pump shaft and a motor shaft is the core presence. For modeling the core section particular precautions need to be taken in consideration. Three diameters need to be defined:

- an external stiffness diameter that takes into account the stiffness of core section including the shaft.
- an internal and an external mass diameter that takes into account the mass and moment of polar inertia of core section.

Stiffness diameter

Supplier needs to provide the torsional stiffness K_t of a rotor segment (i.e., between drive end side and center line of core, Figure 12). Then, it is necessary to know the torsional stiffness of the segment before core (segment A). It is possible to acquire this data from a rotordynamic software, such as Madyn 2000, or by formula. For each section the torsional stiffness (K_i) can be calculated as:

$$K_i = \frac{G}{L}\left(\frac{\pi D^4}{32}\right)$$

Where:

Quantity	SI Units
G = torsional modulus	N/m²
L = section axial lenght	M
D = section diameter	m

The reciprocal of the equivalent torsional stiffness for segment A ($K_{eq,A}$) is equal to the sum of all reciprocal torsional stiffness (K_i) of each section.

$$\frac{1}{K_{eq,A}} = \sum_i \frac{1}{K_i}$$

Hence, the resulting torsional stiffness for the core section (segment B) is done by the difference of supplier torsional stiffness and segment A torsional stiffness as follow:

$$K_B = \frac{1}{\left(\frac{1}{K_t} - \frac{1}{K_A}\right)}$$

Known the torsional stiffness, it is possible to define the effective diameter for motor core stiffness (D_B).

$$D_B = \left(\frac{32L}{\pi G} K_B\right)^{1/4}$$

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Mass diameter

Mass and moment of polar inertia equations of a hollow cylinder are used to define an internal (D_i) and an external (D_e) mass diameter of core section (segment C).

$$m_{tot,C} = \rho \cdot L_C \cdot \frac{\pi}{4}\left(D_e^2 - D_i^2\right)$$

$$J_{tot,C} = \frac{1}{2} m_{tot,C}\left(\frac{D_e^2}{4} + \frac{D_i^2}{4}\right)$$

Where:

Quantity	SI Units
m_{tot} = total mass (segment C)	kg
J_{tot} = total moment of polar inertia (segment C)	kg m^2
L_c = axial length (segment C)	m
D_e = external mass diameter	m
D_i =internal mass diameter	m

- m_{tot} is the sum of shaft and motor core mass of segment C.

$$m_{tot,C} = m_{sh,C} + m_{core}$$

- J_{tot} is the sum of shaft and motor core moment of polar inertia of segment C.

$$J_{tot,C} = J_{sh,C} + J_{core}$$

Motor core mass and moment of polar inertia needs to be provided by the supplier, while shaft mass and moment of polar inertia can be calculated by using mass and moment of polar inertia of a cylinder.

$$m_{sh,C} = \rho \cdot L_C \cdot \frac{\pi}{4} D_{sh}^2$$

$$J_{sh,C} = \frac{1}{2} m_{sh,C} \frac{D_e^2}{4}$$

Known the mass and moment of polar inertia of segment C, internal and external mass diameter can be obtained from equations above:

$$D_e = 2\sqrt{\frac{J_{tot,C}}{m} + \frac{m_{tot,C}}{2\pi\rho L_C}}$$

$$D_i = 2\sqrt{\frac{J_{tot,C}}{m} - \frac{m_{tot,C}}{2\pi\rho L_C}}$$

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In summary, the supplier needs to provide:

- Shaft geometry: length and diameter of each section;
- Material properties (E, G, v, ρ …)
- Torsional stiffness of rotor between drive end side and center line of core;
- Mass and moment of polar inertia of motor core;
- Mass and moment of polar inertia of other parts mounted on motor (i.e., fan).

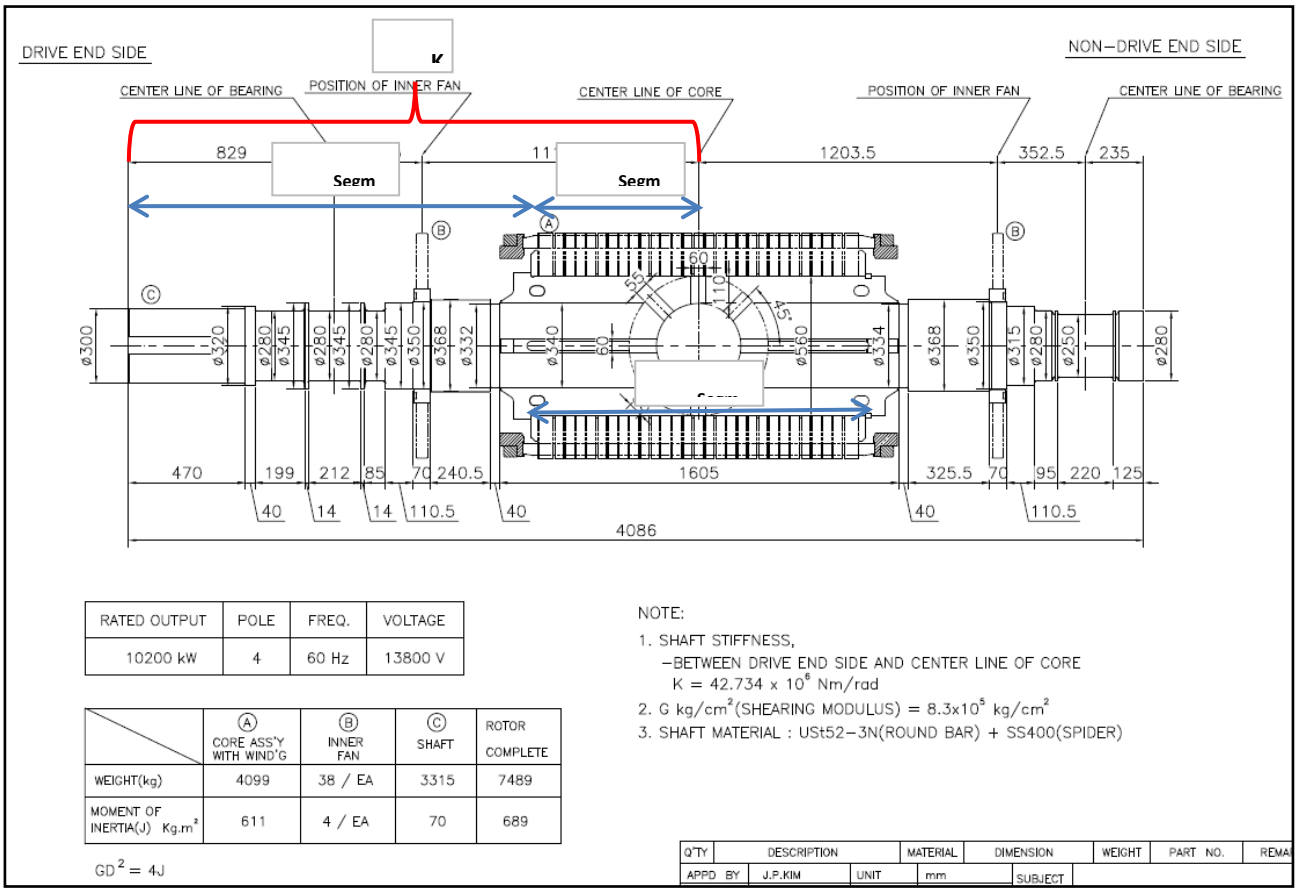

Figure 12. Example of electric rotor

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Figure 13 is an example of electric motor shaft. In the core section three diameter have been used:

- "ODs main" as the stiffness diameter;
- "ODm main" as the external mass diameter;
- "IDm main" as the internal mass diameter.

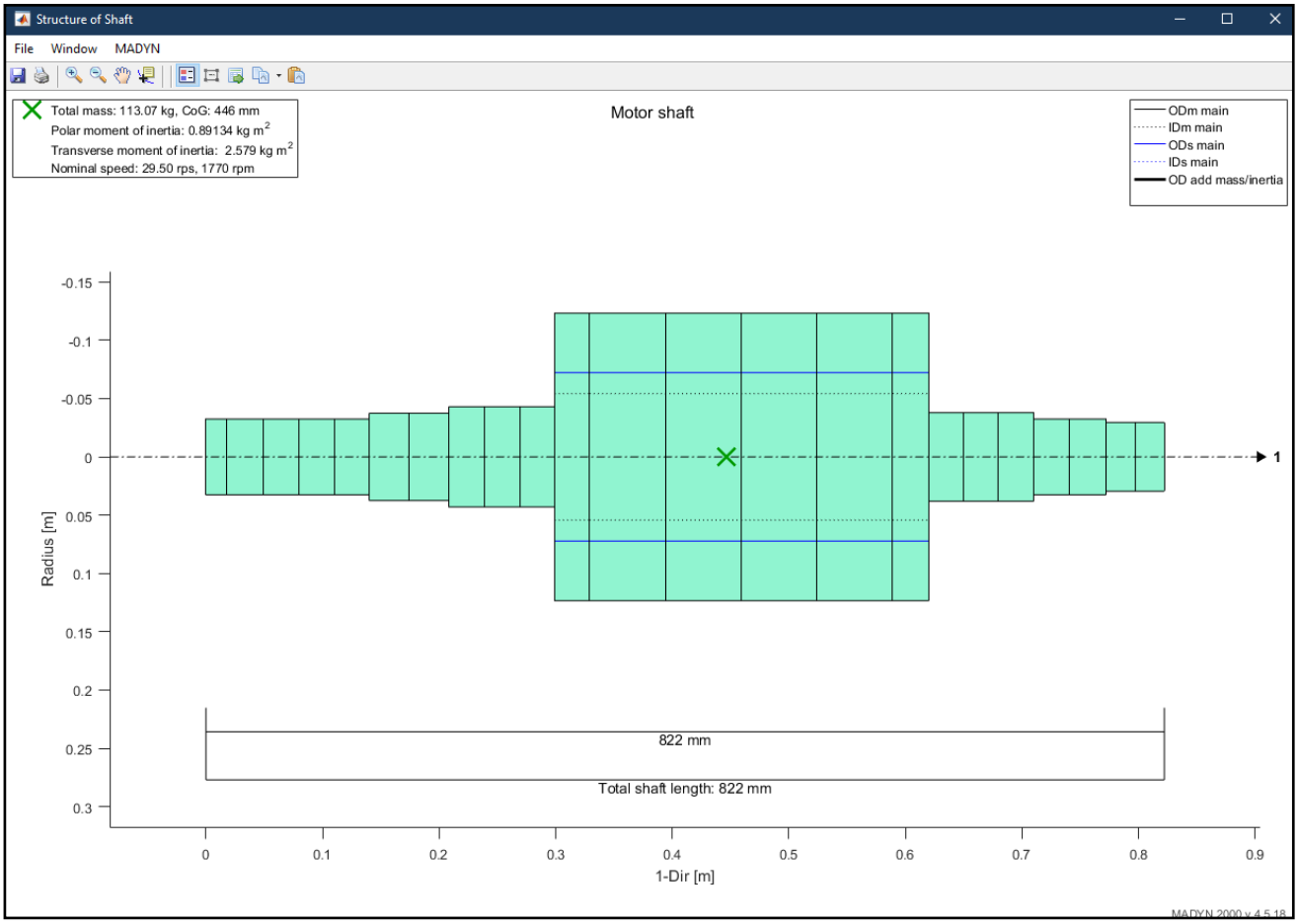

Figure 13. Example of motor shaft.

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2.4. Madyn 2000, Connection objects

Torque transmission can happen trough different ways. The most common are:

- Flexible or rigid coupling;
- Hydraulic coupling;
- Gearbox.

Unless a hydraulic coupling is used to transmit torque, rotors need to be mechanically connected.

Flexible and rigid coupling can be modelled as a torsional stiffness and two lumped masses. These masses takes into account the hub and half spacer inertial properties and should be positioned in its center of gravity.

Usually, flexible coupling inertial properties and torsional stiffness are provided by the supplier. If some data are missing, a CAD model can be used. For example, in Figure 14, inertial properties and center of gravity are not directly provided. Coupling half weight can be obtained by summing each component while, moment of inertia and center of gravity can be obtained with a CAD model.

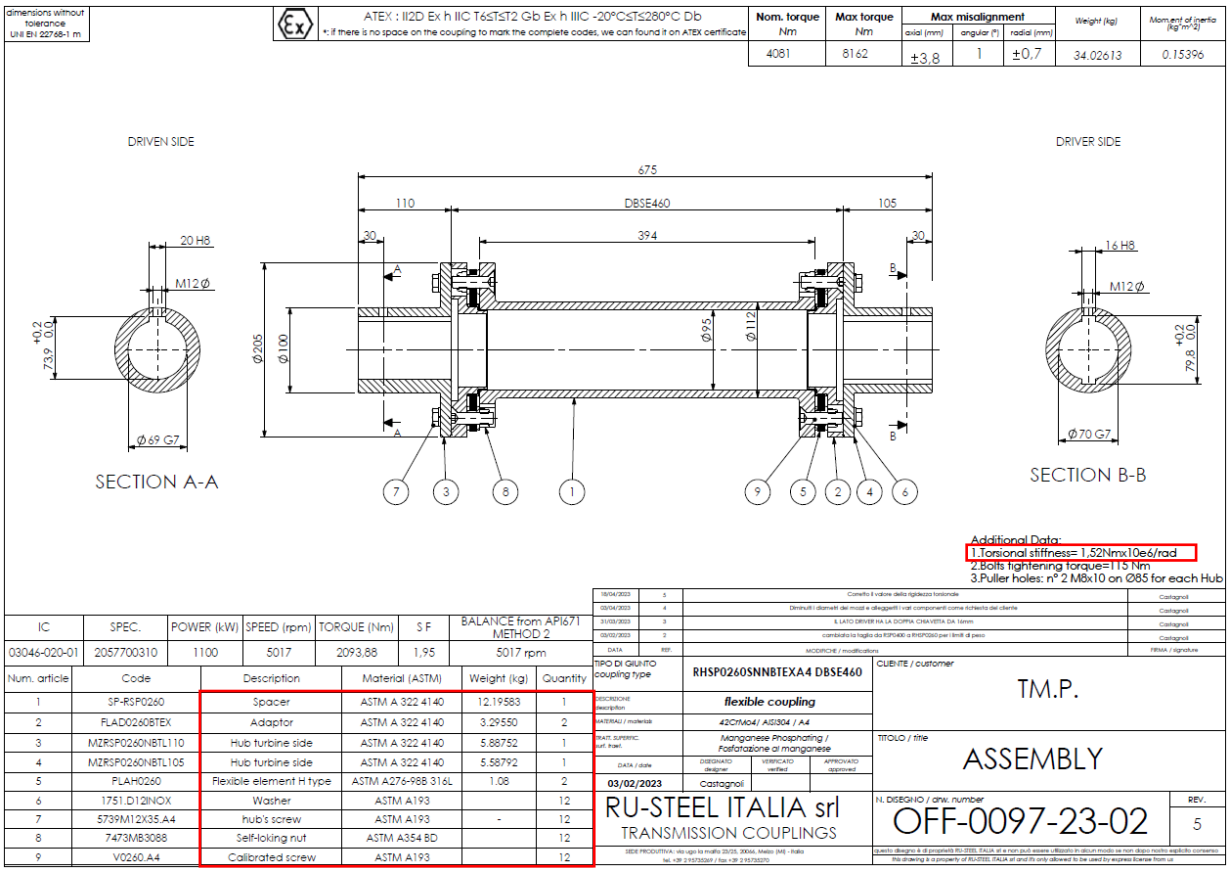

Figure 14. Example of flexible coupling description.

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Rigid couplings are mainly used in vertical pumps. They are internally made so, there is not a general description. Inertial properties and torsional stiffness can be obtained by formulas of a hollow cylinder:

- Torsional stiffness.

$$k_t = GJ/L^* \ [N \cdot mm/rad]$$

- Weight.

$$M = \rho\frac{\pi}{4}\left(D_e^2 - D_i^2\right)L_{tot} \ [kg]$$

- Moment of polar inertia.

$$J_p = \frac{1}{2}M\left(\left(\frac{D_e}{2}\right)^2 + \left(\frac{D_i}{2}\right)^2\right) [kg \cdot mm^2]$$

- Moment of diametral inertia.

$$J_d = \frac{1}{12}M\left(3 \cdot \left(\left(\frac{D_e}{2}\right)^2 + \left(\frac{D_i}{2}\right)^2\right) + L_{tot}^2\right) [kg \cdot mm^2]$$

Where:

- $G = shear\ modulus\ [MPa]$
- $J = \frac{\pi}{32} \cdot \left(D_e^4 - D_i^4\right) [mm^4]$
- $L_{tot} = coupling\ lenght$
- $L^* = \frac{4}{6}L_{tot}\ [mm]\ (effective\ lenght - see\ below)$
- $\rho = density\ [kg/mm^3]$

A simpler way to define coupling inertial properties is to use a mass diameter (IDmb and ODmb) in the rotor model as described in *2.2 Madyn 2000, Shaft object*. In this way it is not necessary to create two lumped masses because coupling inertial properties will be already included.

Normally a vendor-supplier torsional stiffness includes an assumed penetration factor (PF). The traditional assumption is that $PF = 1/3$. Stated simply, the shaft is considered unrestrained by the coupling hub for a distance of 1/3 of the length of the fit, starting at the coupling hub edge opposite the shaft end.

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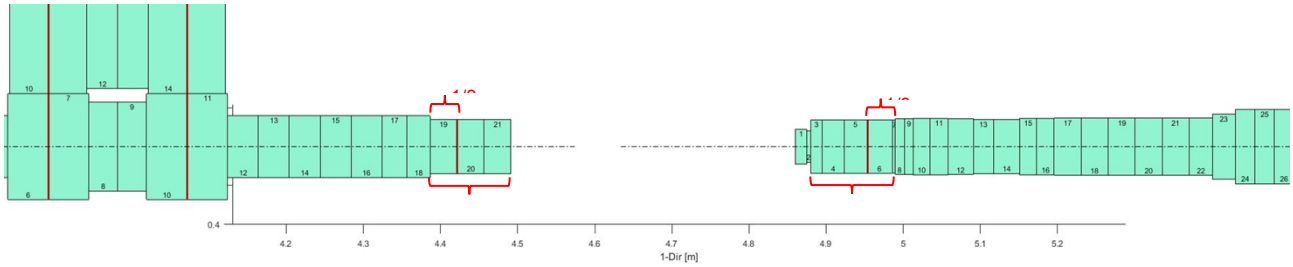

Figure 15. Example of penetration factor (PF).

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In Madyn 2000, rigid and flexible coupling can be modelled with the flexible coupling object (Figure 16).

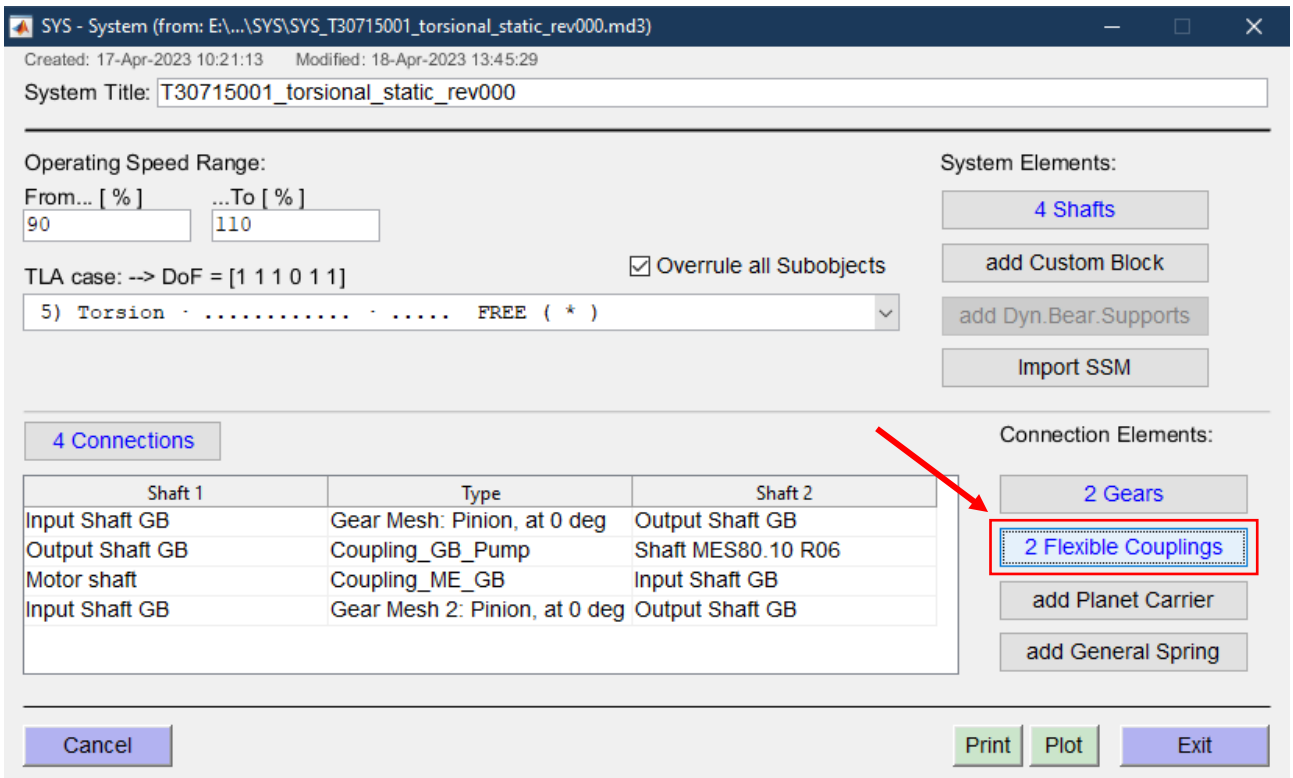

Figure 16. System GUI.

Flexible coupling object inputs (Figure 17) are:

- *Length*. It is the distance between coupling station on the two rotors and is equal to coupling length less two times the penetration factor;
- *Stiffness*. It is the torsional stiffness of the coupling.
- *Masses*. Left and Right define the inertial properties of half coupling on each rotor. Offset defines the position of center of gravity. Beware that it is equal to the center of gravity starting from the hub edge opposite to the shaft end less the penetration factor.
- *Flexible Coupling Connection*. It defines the station to connect and the type of connection used.

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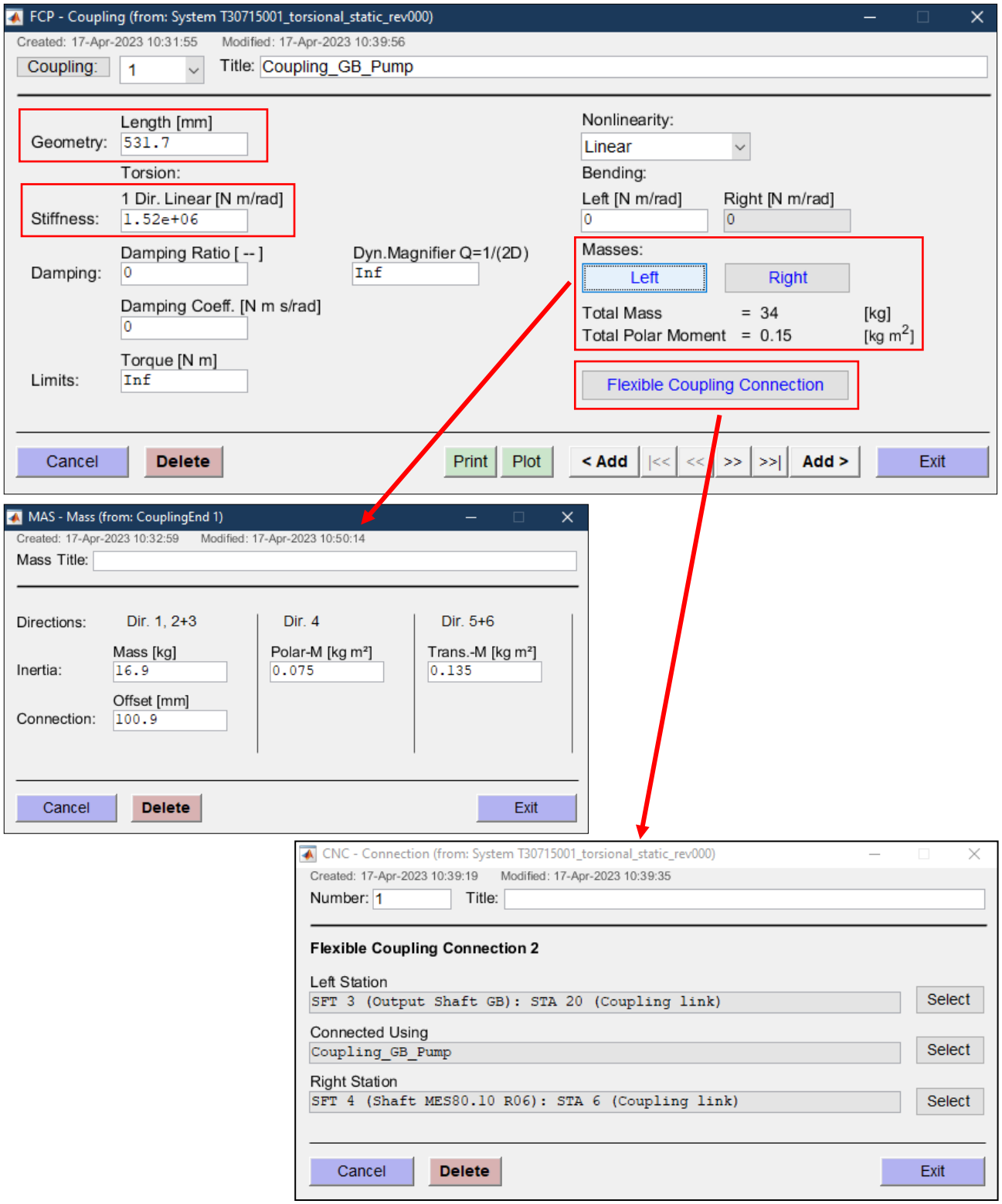

Figure 17. GUI Flexible Coupling Object and Input Data.

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Gearbox can be modeled creating gearbox shafts and connecting them through gear connections. Shafts can be created as explained in *2.2 Madyn 2000, Shaft object*. Supplier needs to provide a description of the shafts (Figure 18). Connections between shafts can be obtained through gear objects (Figure 19).

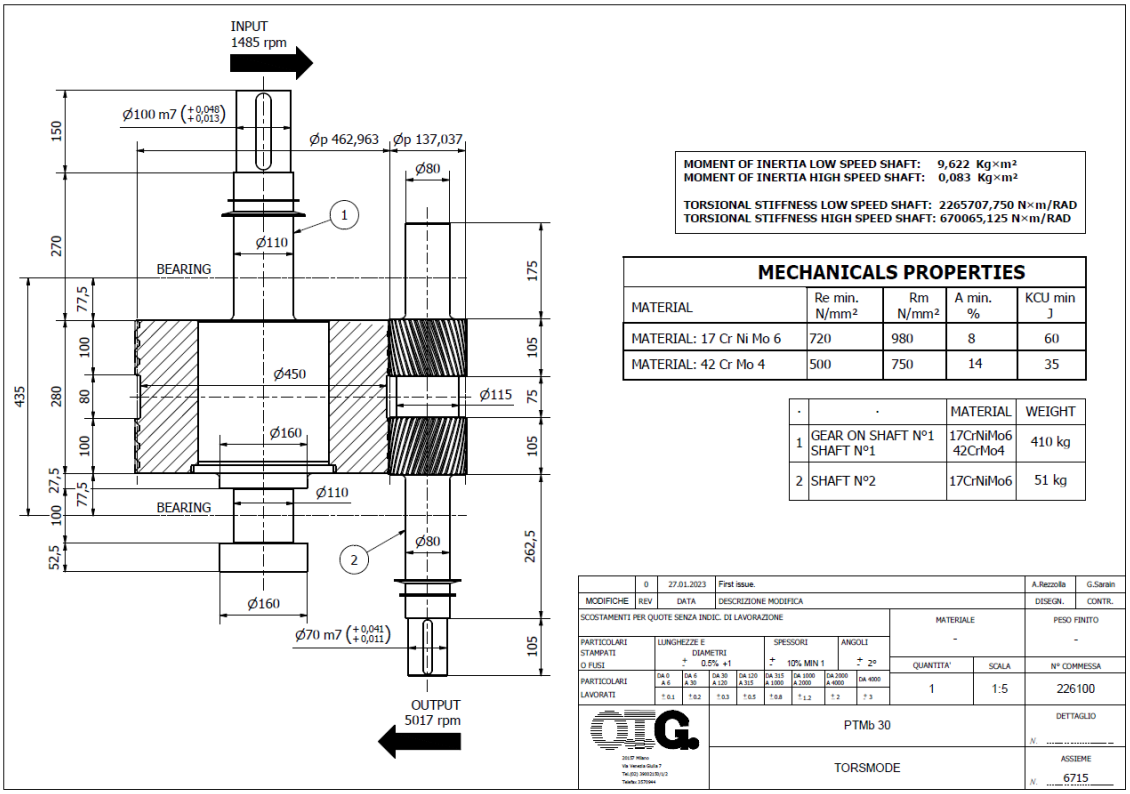

Figure 18. Example of gearbox rotor.

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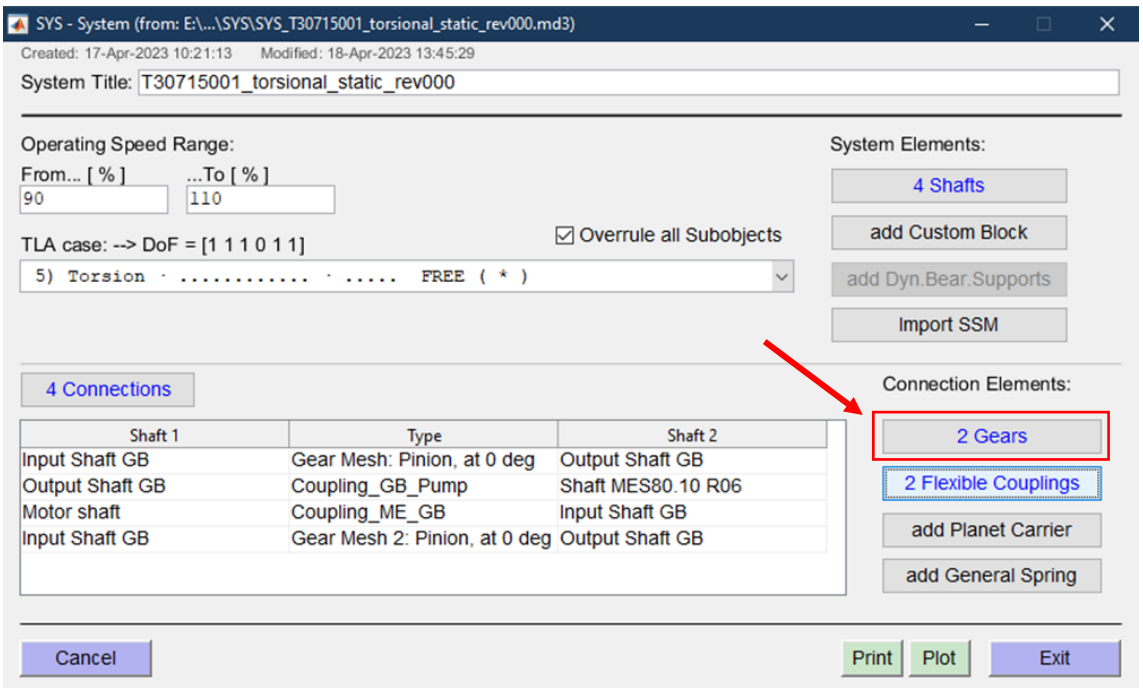

Figure 19. System GUI.

A gear object consists of the gear mesh information, which connect the wheel and one or several pinions. The objects to be connected by the meshes must be part of the system. The main GUI consists of the wheel properties. A GUI to define pinion properties can be opened from the “Pinion” button in this GUI (Figure 20).

Wheel mesh inputs are the wheel mesh radius and mesh type. The mesh can be an outer mesh as for a parallel gear or an inner mesh as for the mesh between planet and annulus of a planetary gear.

Pinion mesh input is the mesh radius. The tooth stiffness (including oil film) is set as default.

The pinion mesh and wheel mesh radii determine the gear ratio.

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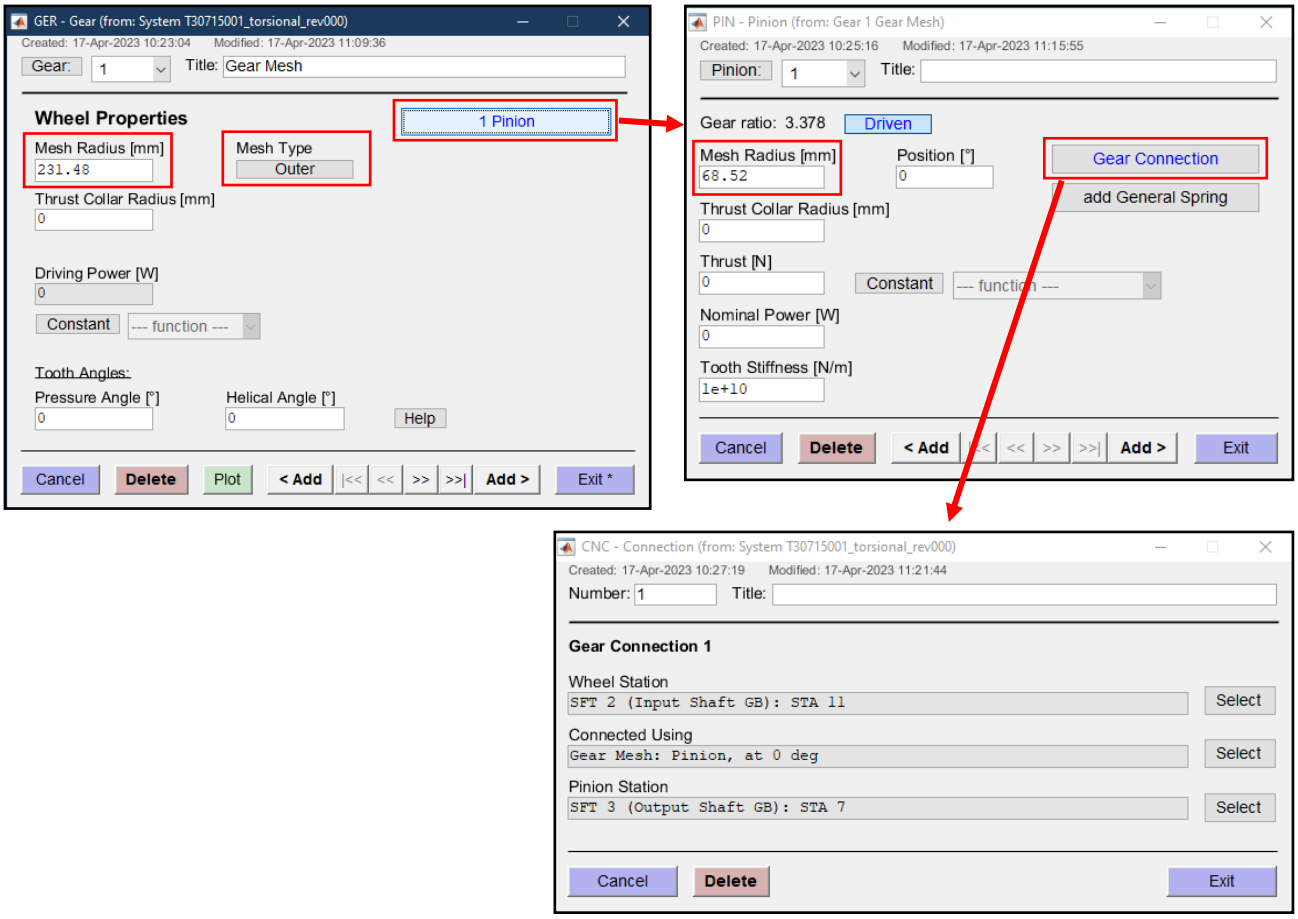

Figure 20. GUI Gearbox Object and Input data.

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2.5. Madyn 2000, Eigenvalue Analysis, Campbell diagram.

The first step in the creation of the Campbell diagram is the calculation of the torsional natural frequencies of the train in study. Figure 21 shows the procedure of eigenvalue calculation in Madyn 2000.

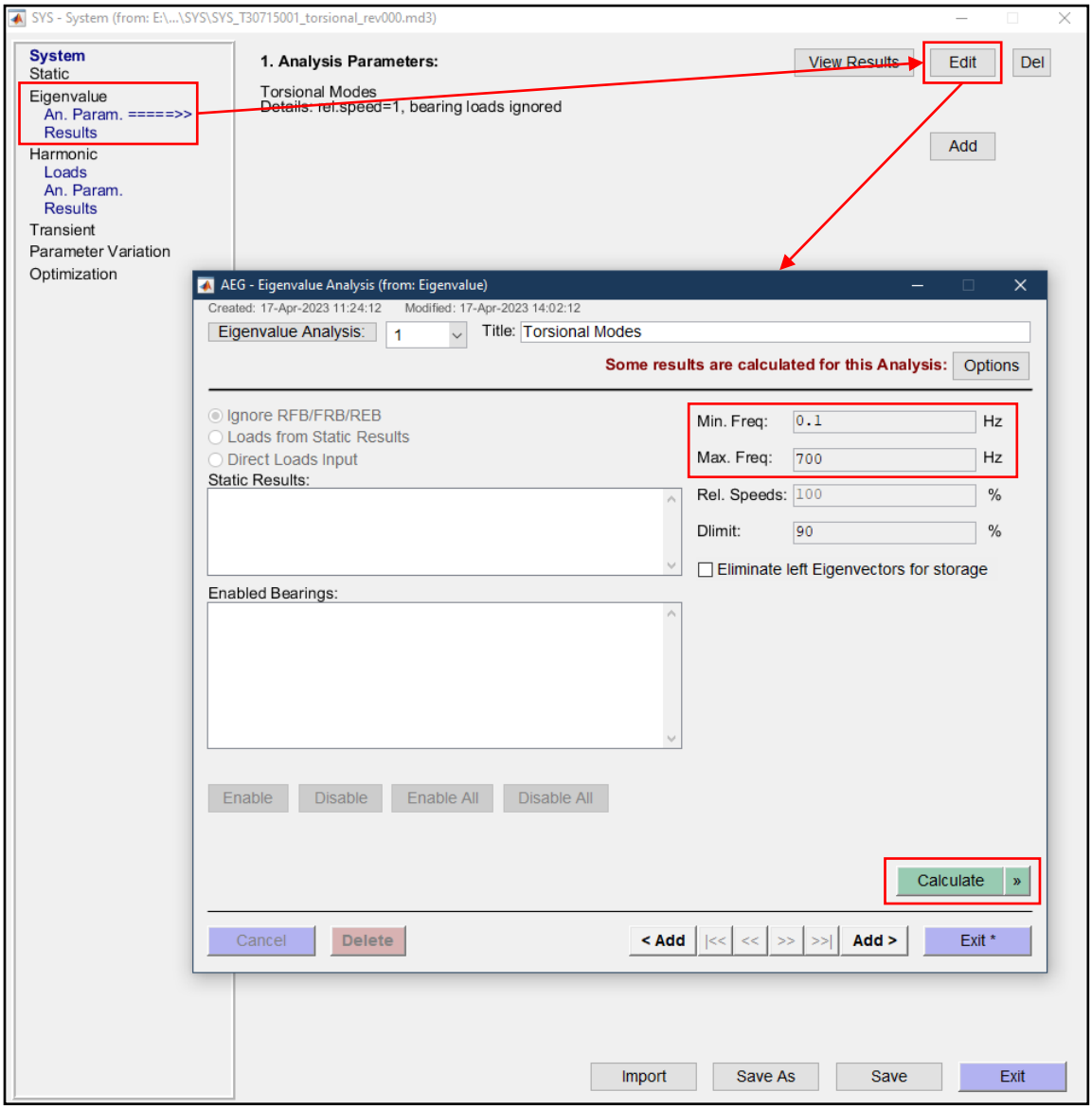

Figure 21. GUI Eigenvalue Analysis

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Minimum Frequency was set a little higher of 0 Hz in order to avoid rigid torsional modes. Rigid modes appear at 0 Hz in unrestrained systems. If there is more than one torsional natural frequency near 0 Hz it is possible there are some errors, such as a wrong value in coupling torsional stiffness. A suggestion is to perform a first run with minimum frequency at 0 Hz to check possible errors, then set a value little higher, such as 0.1 Hz, to neglect rigid mode.

Maximum Frequency should be set higher than:

- $maximum\ rotating\ speed\ \cdot 1.1 \cdot maximum\ order\ excitation\ [Hz]$

For a pump with a rotating speed of 5017 rpm and with maximum excitation order of 7, it is sufficient a maximum frequency of 700 Hz.

Once the analysis has been performed, results can be checked through “Plot” or “Print” command. The main results to be checked are:

- the torsional natural frequencies;
- the associated modes.

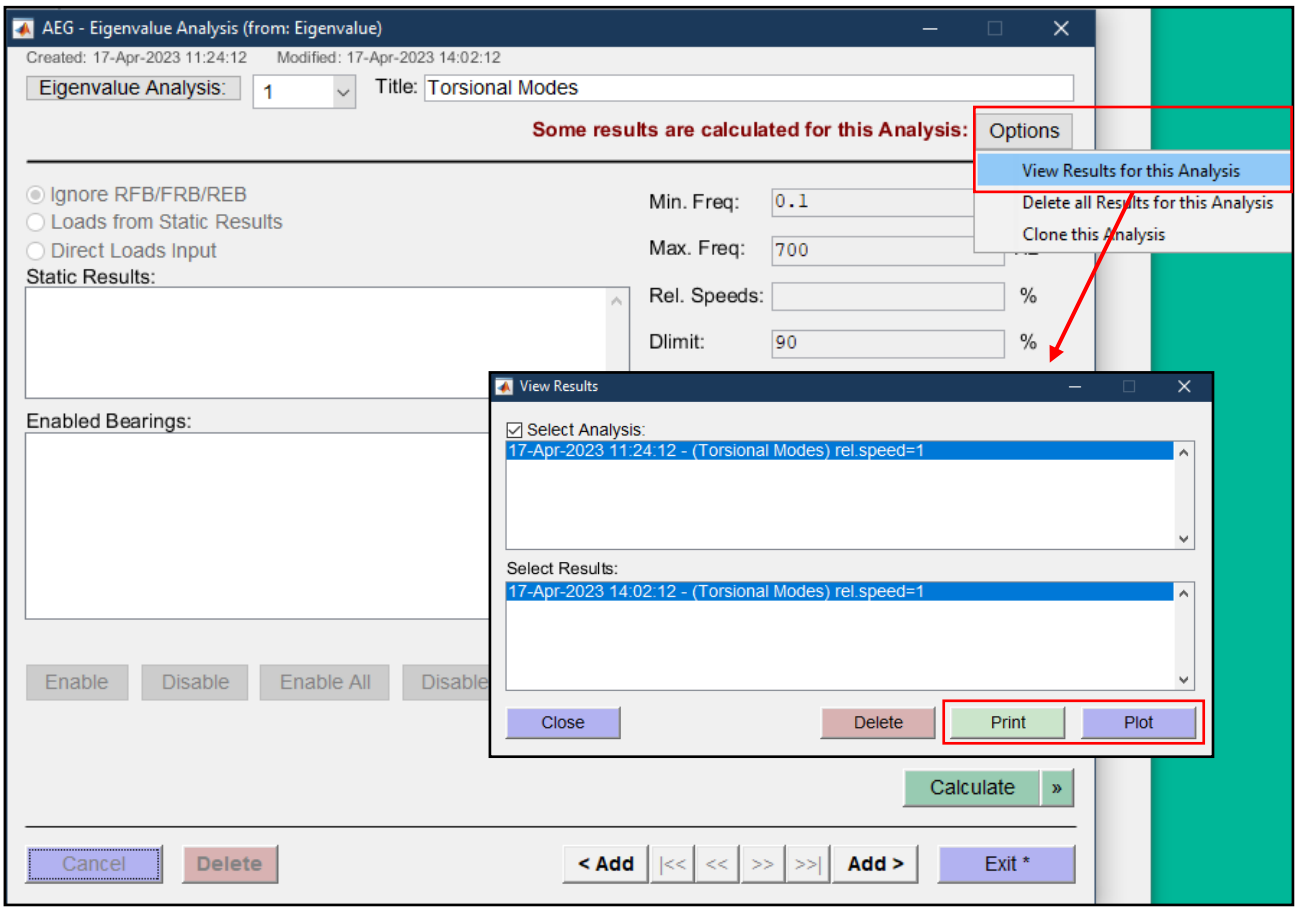

Figure 22. GUI Results

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With the torsional natural frequency values and the help of an excel file (*Mod.407 - Torsional analysis Campbell diagram*), it is possible to create the Campbell diagram (Figure 23).

- Torsional natural frequencies are plotted as horizontal lines (green lines).
- Operating speed range is designated by vertical lines (blue lines).
- The upward sloping lines are harmonics of speed that represent the system's potential excitations (red lines). If excitation sources are not a function of speed, such as line frequency, they are represented as straight lines (violet lines).
- Intersections between excitation lines and natural frequency lines are referred to as interference points (red dots). They are indicators of potential resonances if they occur within the operating speed range.

If there are some critical speeds inside the operating speed range, a harmonic analysis (steady state response analysis) and a fatigue verification need to be performed.

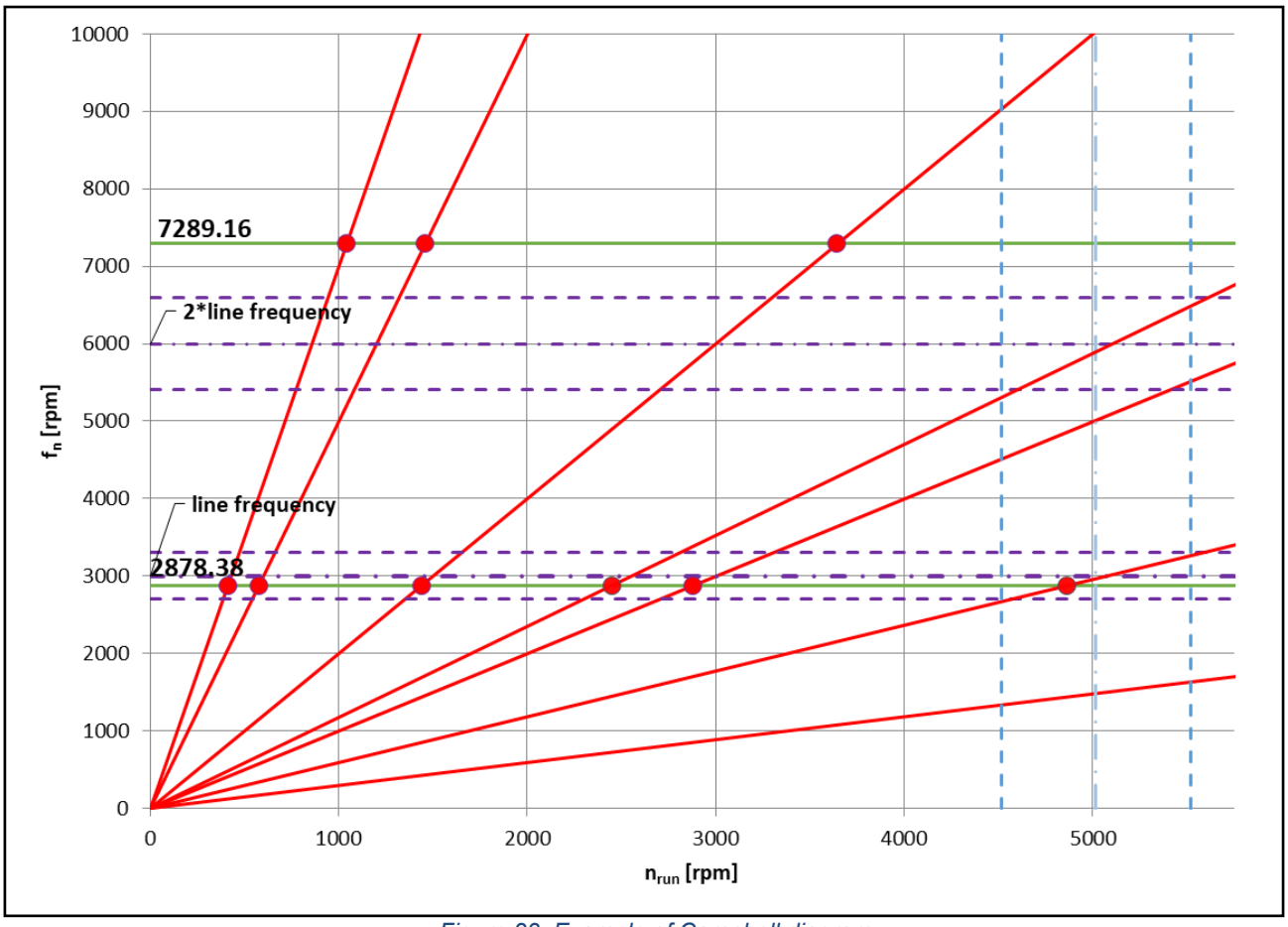

Figure 23. Example of Campbell diagram.

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2.6. Madyn 2000, Harmonic Analysis.

The first step in a steady state response analysis is the definition of the loads. Figure 24 shows the type of loads mainly used for a harmonic analysis:

- The **Harmonic Force** used for synchronous excitations, such as excitation at rotating speed (1X) and impressed voltage VFD excitation.
- The **Integer and Non-integer harmonic Excitation** used for harmonics of rotating speed, such as blade pass, and electric excitation linked to line frequency.

  Note. This type of excitation can be used to also define excitations at synchronous speed.

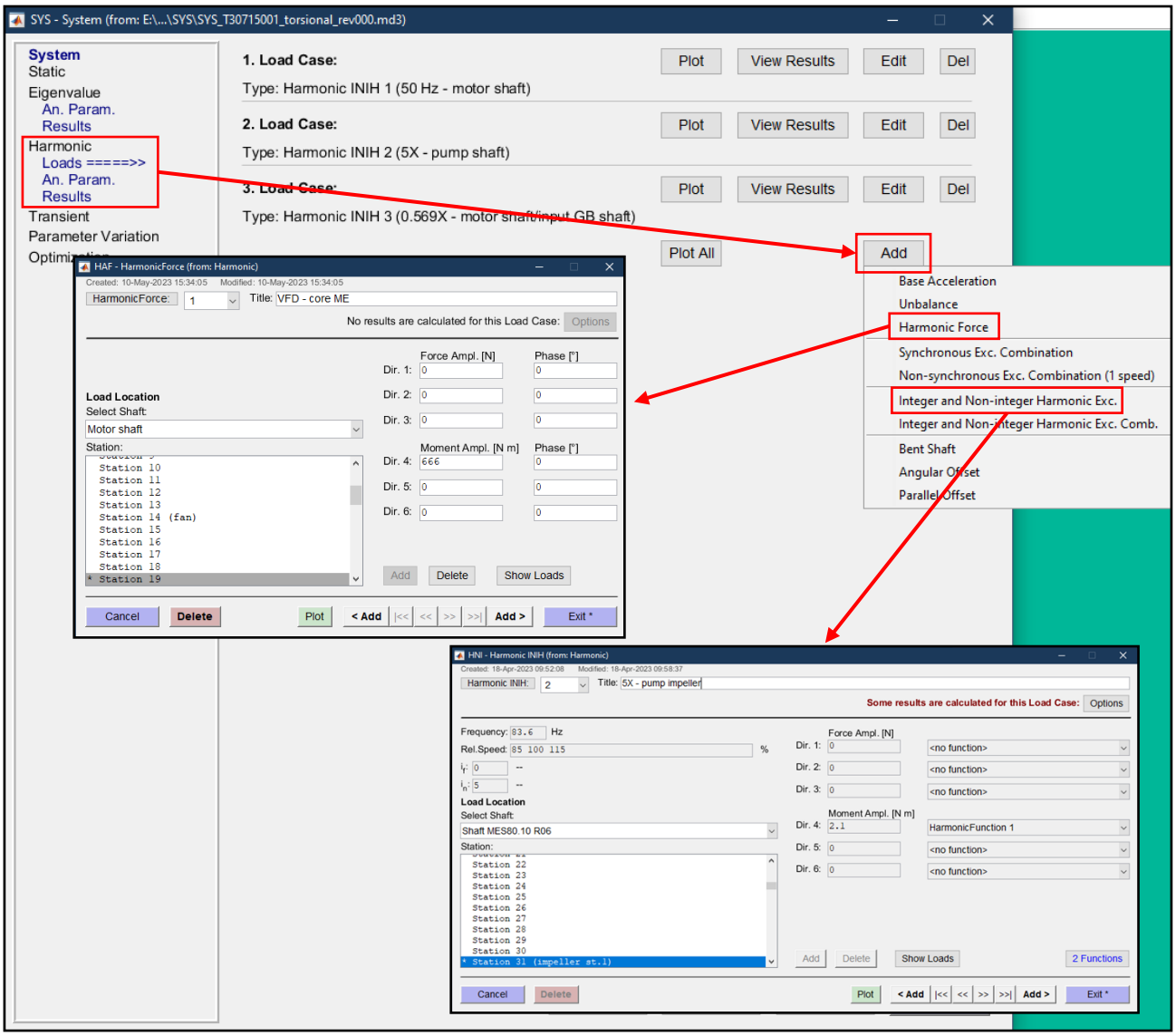

Figure 24. GUI Harmonic Loads

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Harmonic Forces can be applied at shaft stations and mass centers of gravity as well. They are defined by force and moment amplitudes for all 6 degrees of freedom (for these cases define a moment amplitude in direction 4) and a time phase angle ϕ for each component.

The excitation function for a harmonic force is as follows:

$$F = \hat{F}\cos(\omega t + \phi)$$

with ω as the excitation frequency in [rad/s].

After selecting the load location (i.e., station 19 of motor shaft), amplitude $\hat{F}$ and phase ϕ can be defined activating the load boxes with the "Add" button (Figure 25).

All excitations should be applied using the worst-case scenario of load location and phase for the mode shape in study.

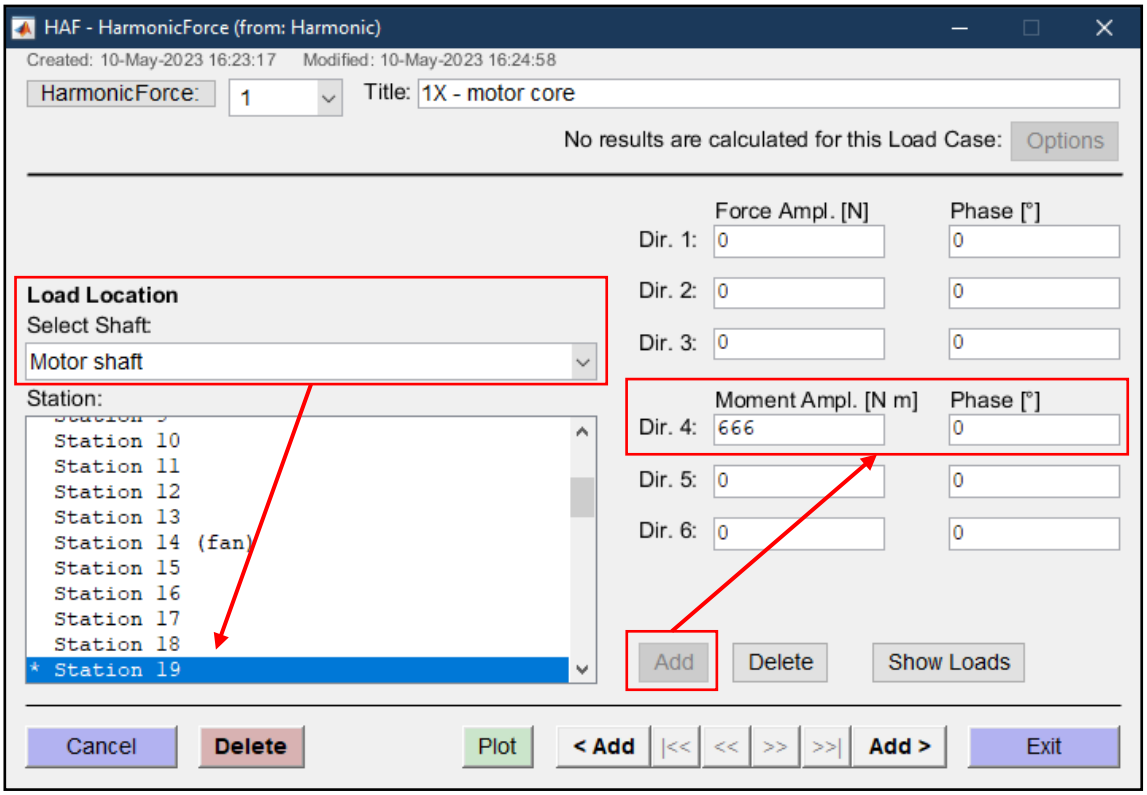

Figure 25. GUI Harmonic Force

The excitation frequency for harmonic forces ω can be equal to the rotational speeds Ω if the check box "Synchronous Excitation" in the Analysis Parameters will be activated (Figure 26).

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Before launching the load cases, it is important to define:

- the relative speed range. For synchronous excitations the relative speed range goes from 90% (or 85%) to 110% (or 115%). VFD cover a frequency range up to 1000 Hz. Hence, the starting value is near 0% (i.e., 0.1%), while the end value is equal to $1000\ Hz/(speed/60)$;
- the structural damping. In general, the assumption is a structural damping of 1%.

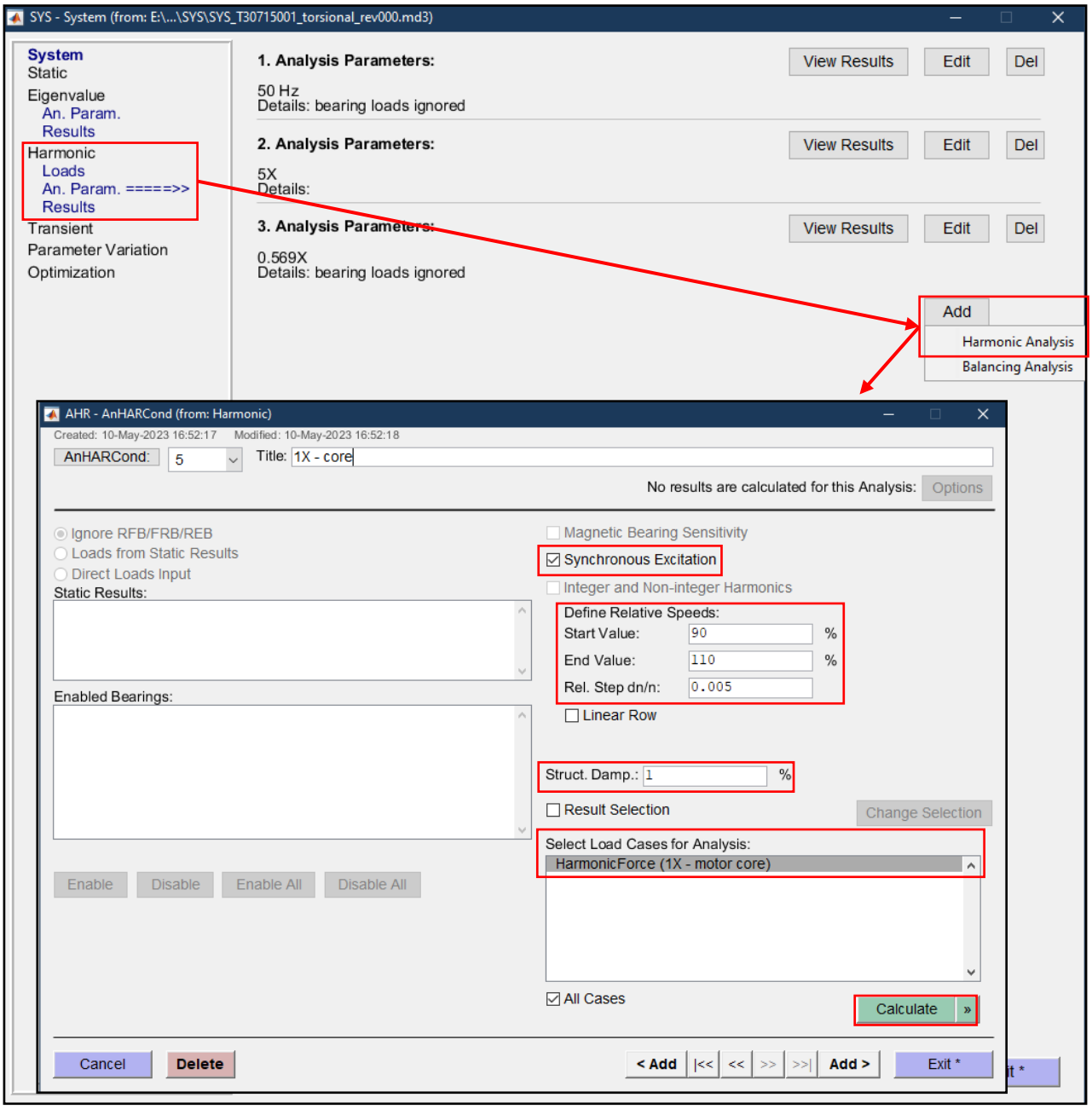

Figure 26. Harmonic Analysis. Synchronous excitation.

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Integer and Non-integer harmonic Excitation (INIH) objects allow defining factors if and in by which the speed n and the frequency f are multiplied and combined to define an excitation frequency according to the formula:

$$f_e = i_n n + i_f f$$

Torque excitations with frequencies multiples of rotating speed can be defined by putting $i_n = x$ & $i_f = 0$ where x is the harmonic order (Figure 27). This is an easy way to define excitations such as misalignment, blade pass, …. Excitations proportional to a frequency such as line frequency can be defined by putting $i_n = 0$ & $i_f = x$ where x is the order of the frequency relative to the frequency defined (Figure 28). Also, floating point numbers such as 0.5 can be defined as INIH loads.

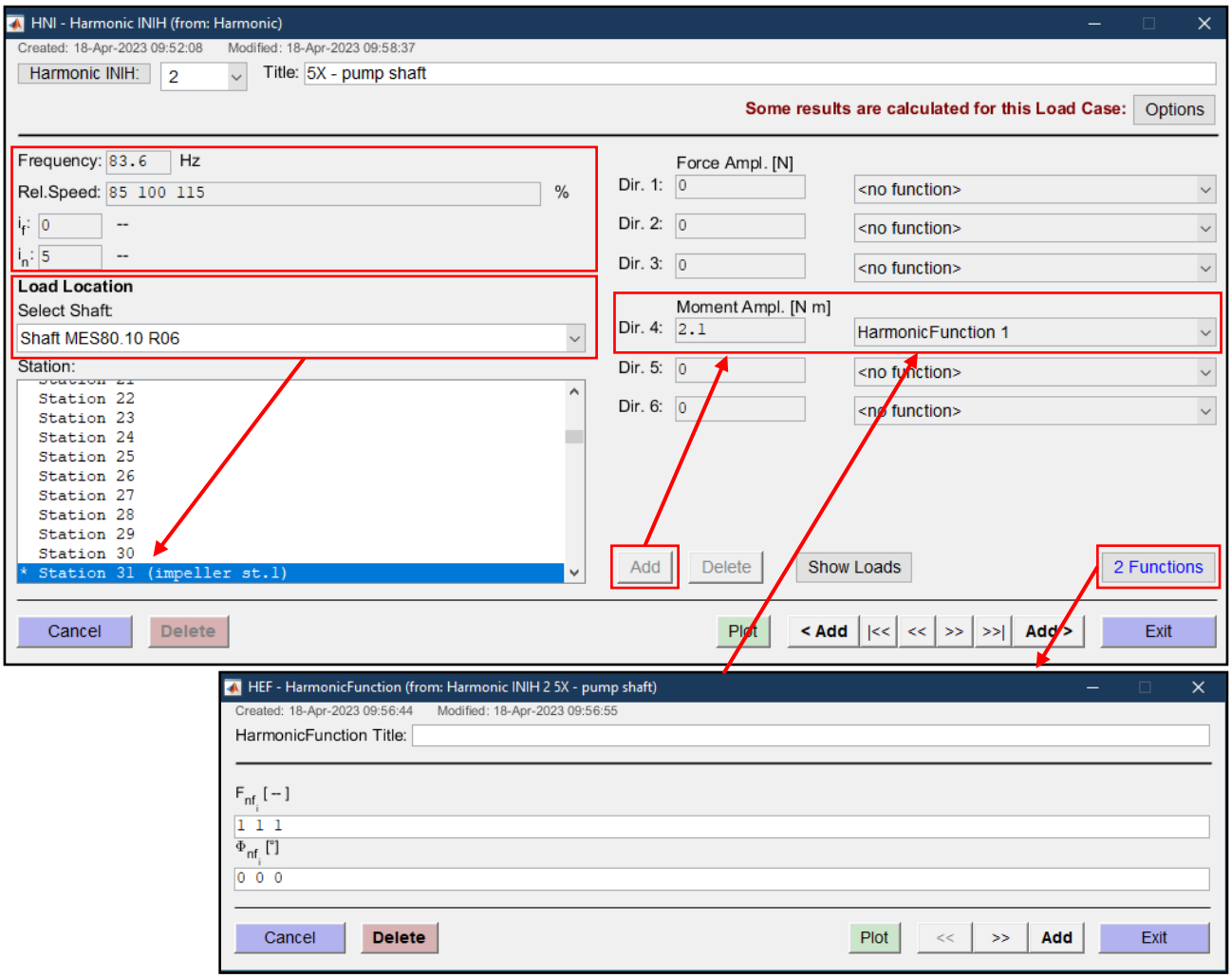

Figure 27. GUI INIH Load. Example of excitation proportional to rotating speed.

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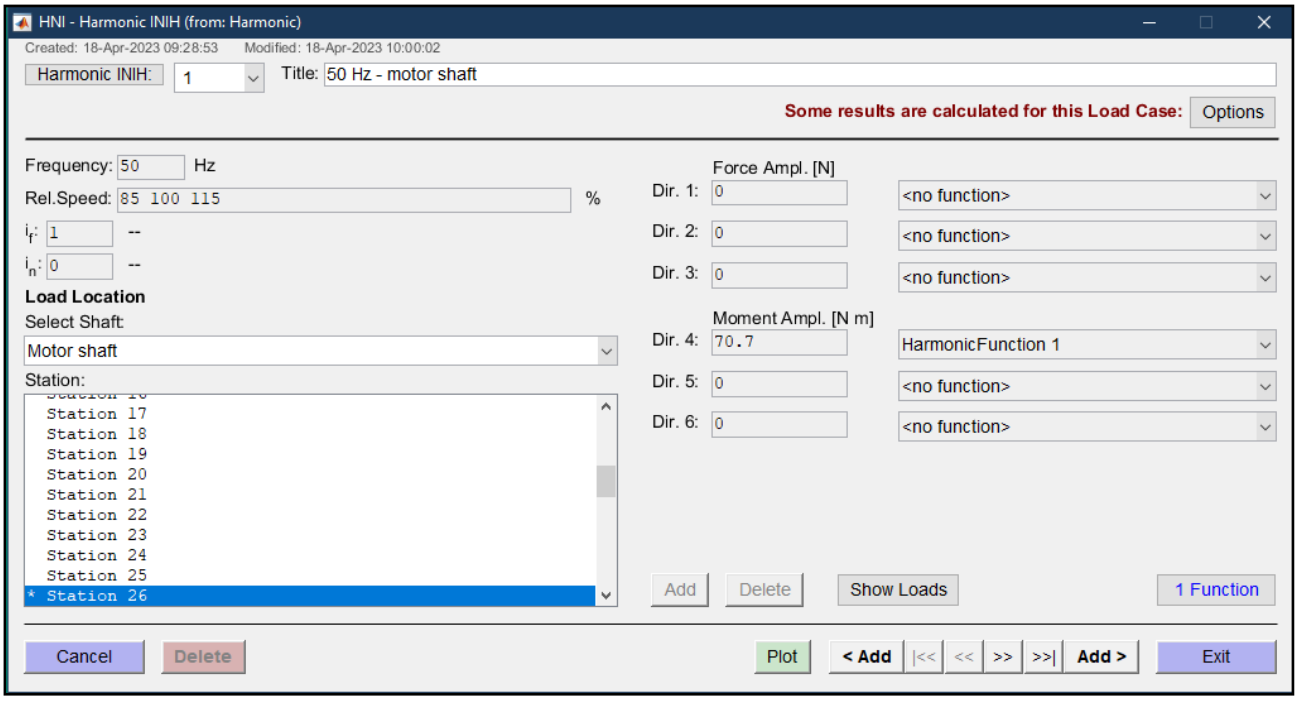

Figure 28. GUI INIH Load. Example of excitation proportional to excitation.

The defined forces and moments may vary with the speed. They can be multiplied with a speed function f(n). The speed function for each of the defined speeds in the load GUI can be input in a separate window, which is invoked by the button “add Function” (Figure 27).

The definition of load locations and load amplitude is the same as the harmonic force.

With the INIH load, in the Analysis Parameters, the check box “Integer and Non-integer Harmonics” should be activated (Figure 29).

As for the harmonic force, before launching the load cases, it is important to define:

- the relative speed range that goes from 90% (or 85%) to 110% (or 115%);
- the structural damping that, in general, is assumed equal to 1%.

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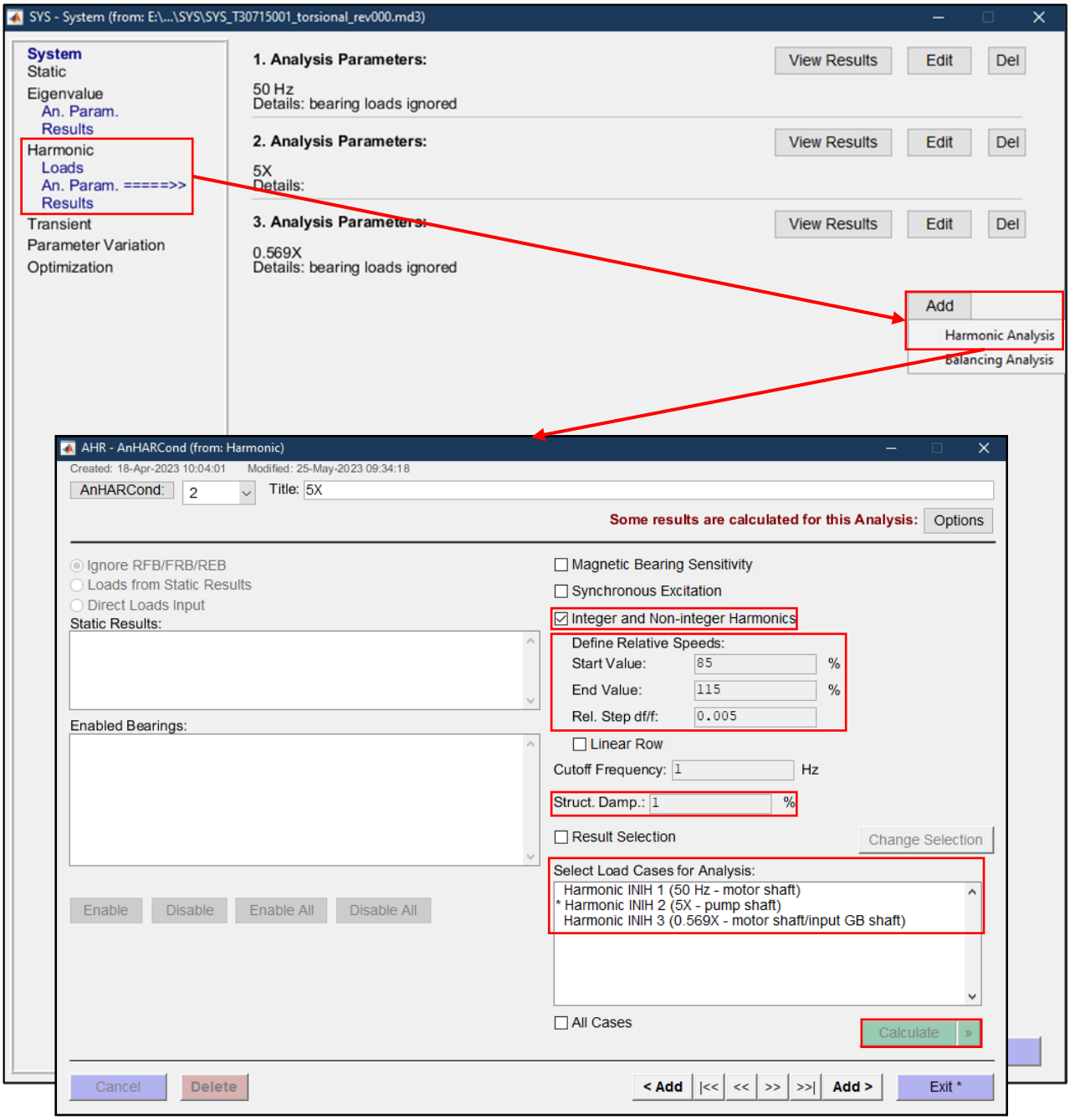

Figure 29. Harmonic analysis. INIH excitation.

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2.7. Fatigue Analysis, Goodman diagram.

The dynamic torsional torques and stresses generated from the steady-stated response analysis should be used to evaluate mechanical integrity of the system. Shafts encounter a fluctuating stress that is a combination of static stress due to average transmitted torque and cyclic stress due to torsional vibration.

Figure 30 gives a convenient graphical representation of various combinations of mean and alternating stress in relation to criteria both for yielding and for various fatigue lives. It is often called a constant-life fatigue diagram because it has lines corresponding to a constant 10^6-cycle (or "infinite") life, constant 10^5-cycle life, and so forth.

The horizontal axis (σ_a = 0) corresponds to static loading. Yield and ultimate strengths are plotted at points A and B. If the mean stress is zero and the alternating stress is equal to S_y (point A″), the stress fluctuates between $+S_y$ and $-S_y$. All points along the line AA″ correspond to fluctuations having a tensile peak of S_y.

Points C, D, E and F correspond to a fatigue life of different number of cycles. Connecting these points with B gives estimated lines of constant life. This empirical procedure for obtaining constant-life lines is credited to Goodman.

Any torsional mean stress is considered positive, so omit left half of diagram. Instead of S_y and S_u, use $C_L S_y$ and $0.8S_u$ (Table 2) for steel.

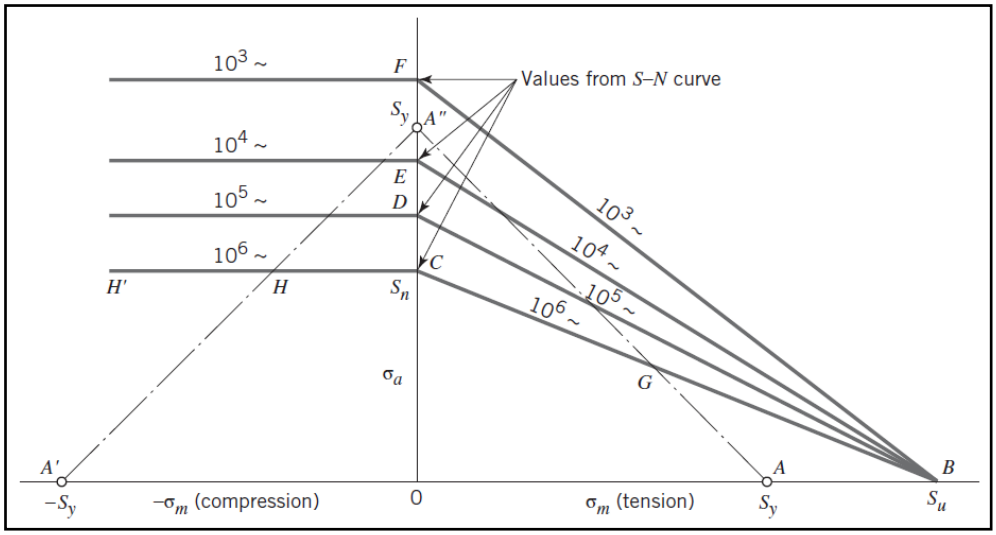

Figure 30. Fatigue diagram.

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The intensity of reversed stress causing failure after a given number of cycles is called the **fatigue strength** corresponding to that number of loading cycles. Numerous tests have established that ferrous materials have an **endurance limit**, defined as the highest level of alternating stress that can be withstood indefinitely without failure (**S_n**). It is designated as **S'_n** where the prime indicates the special case of the standard test on a specimen.

For steel, lacking better data, it can be assumed:

$$S_n^{'} = 0.5S_u$$

For ductile metals, the endurance limit (or long-life endurance strength) in *reversed torsion* is about 58% of endurance limit (or long-life endurance strength) in *reversed bending*. This is taken into account by multiplying the basic endurance limit S'_n by a **load factor C_L** of 0.58.

Up to this point, all discussions of fatigue strength have assumed the surface to have a special "mirror polish" finish. Normal commercial surface finishes usually have localized points of greater fatigue vulnerability: hence the commercially finished parts have lower fatigue strengths. The amount of "surface damage" caused by the commercial processes depends not only on the process but also on the susceptibility of the material to damage. Figure 31 gives estimated values of **surface factor (C_S)** for various finishes applied to steels of various hardnesses.

A comparison of stress gradients shows that large specimens in torsion do not have the same favourable gradients as the standard 10 mm specimen. Experiments show that if the diameter is increased to much more than 10 mm, most of the beneficial gradient effect is lost. Hence, parts that are more than 10 mm in diameter and that are subjected to reversed torsion should carry a **gradient factor (C_G)** of 0.9. Parts with sections larger than about 50 mm equivalent diameter are usually found to have lower endurance limits than those computed using the gradient factors recommended. This is due in part to metallurgical factors such as hardenability, for the interior of large-section parts is usually metallurgically different from the surface metal. The extent to which the endurance limit of very large parts is reduced varies substantially. A very rough guide for the values sometimes used is given in Table 2.

Generalized empirical factors were given for use when data are not available. These factors can be applied with greatest confidence to steel parts because most of the data on which they are based came from testing steel specimens.

Five of these factors are involved in the estimate for endurance limit:

$$S_n = S_n^{'} C_L C_G C_S C_T C_R$$

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The **temperature factor (C_T)** accounts for the fact that the strength of a material decreases with increased temperature, and the **reliability factor (C_R)** acknowledges that a more reliable (above 50%) estimate of endurance limit requires using a lower value of endurance limit.

Table 2 gives a summary of all factors used for estimating the fatigue strength of ductile materials (when subjected to completely reversed loading). It serves as a convenient reference for solving problems.

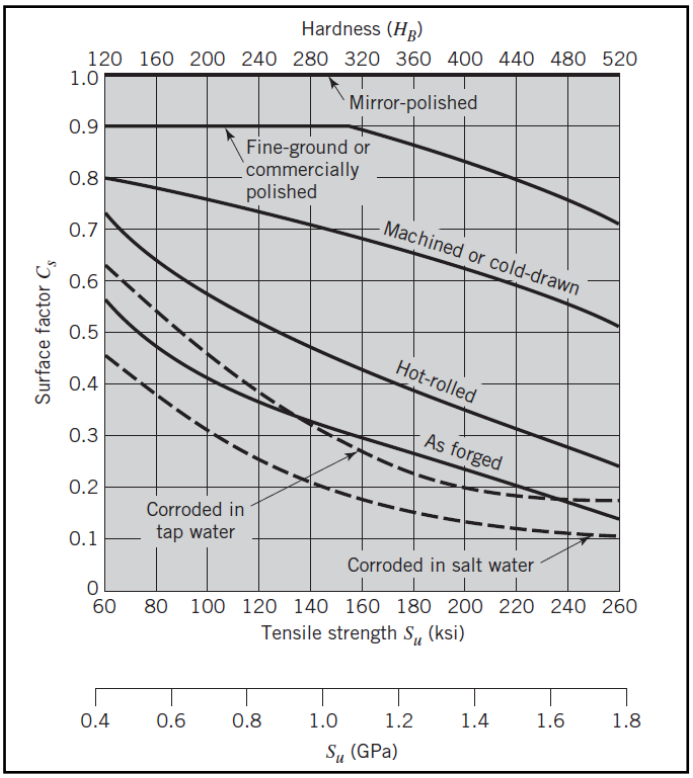

Figure 31. Reduction in endurance owing to surface finish – steel parts

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Table 2. Fatigue Strenght factors for ductile materials.

Torsion

10^6-cycle strenght

Endurance limit: $S_n = S_n^{'} C_L C_G C_S C_T C_R$

where $S_n^{'} = 0.5 S_u$ *for steel, lacking better data.*

C_L	*(load factor)*		0.58
C_G	*(gradient factor)*	diameter < 10 mm	1.0
		10 mm < diameter < 50 mm	0.9
		50 mm < diameter < 100 mm	0.8
		100 mm < diameter < 150 mm	0.7
C_S	*(surface factor)*	see Figure 31	
C_T	*(temperature factor)*	T ≤ 450 °C	1.0
		450 °C ≤ T ≤ 550 °C	$1 - (0.00576T - 2.5856)$
C_R	*(reliability factor)*	50% reliability	1.0
		90% reliability	0.897
		95% reliability	0.868
		99% reliability	0.814
		99.9% reliability	0.753 (suggested)

10^3-cycle strength

Torsional loads: $S_f = 0.9 S_{us} C_T$

where S_{us} *is the ultimate shear strength.*

$$\begin{cases} S_{us} \approx 0.8 S_u \; for \; steel \\ S_{us} \approx 0.7 S_u \; for \; other \; ductile \; metals \end{cases}$$

For torsional analysis omit left half of diagram (any mean torsional stress is considered positive); for steel, instead of S_y and S_u, use $C_L S_y$ and $0.8 S_u$ (Table 2) for steel.

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Stresses should be increased by an appropriate **fatigue stress concentration factor (K_f)** to account for the presence of notch in the shaft. Test show that K_f is often less than geometric factor K_t.

To be precautionary it is assumed:

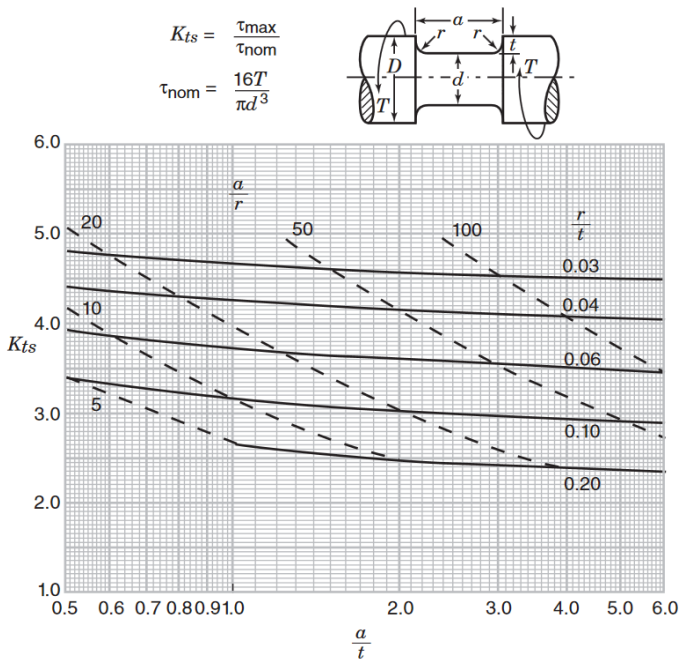
$$K_f = K_t$$

Typical notches present in a shaft are:

- Grooves;
- Keyway;
- Fillet radius.

Table 3 gives a general indication of K_f for the notches listed above.

Table 3. Fatigue stress concentration factors.

Groove. Stress concentration factor varies a lot from the geometry, so a constant value can't be assumed.	 **Chart 2.52** Stress concentration factors K_{ts} for a shaft in torsion with flat-bottom groove (Rubenchik 1975; Grayley 1979; ESDU 1981).

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Keyway.

For diameter less than 165 mm, it is reasonable to assume:

- $K_{tsA} = 1.7$
- $K_{tsB} = 3.0$

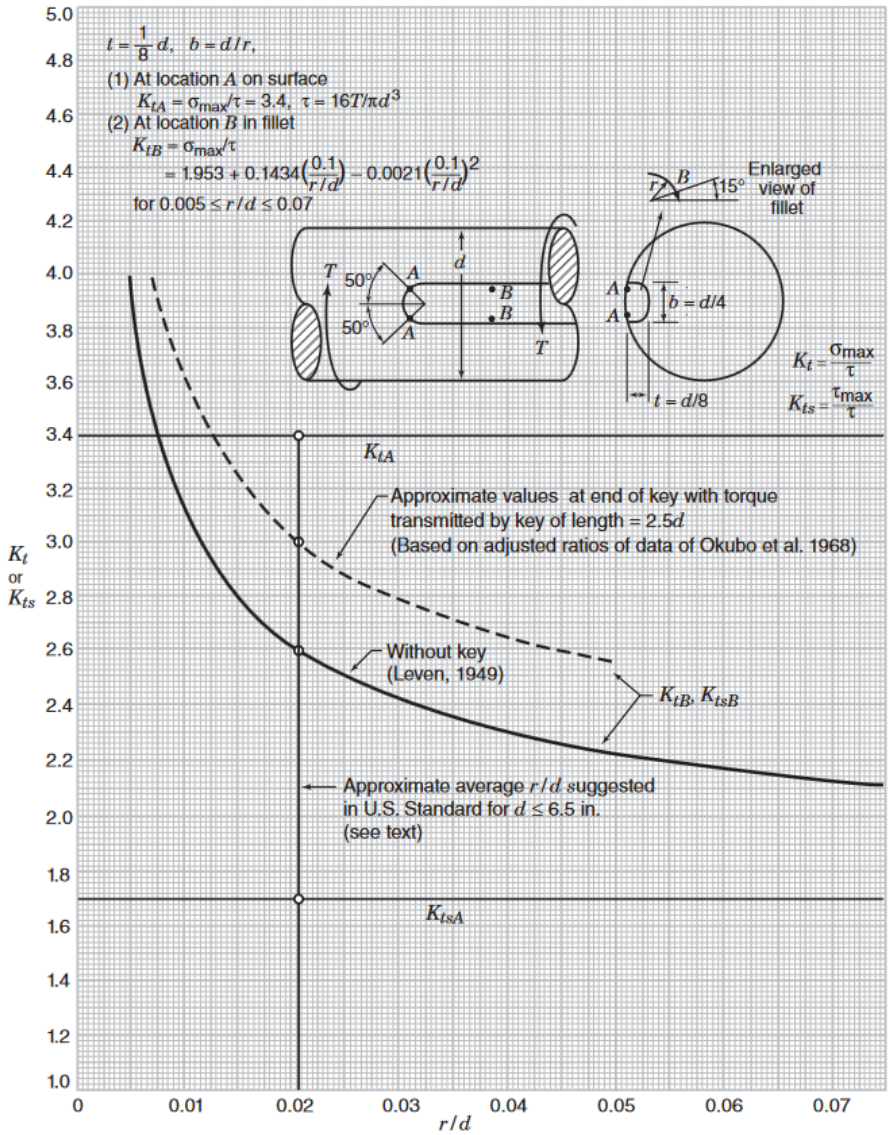

Chart 5.2 Stress concentration factors K_t, K_{ts} for a torsion shaft with a semicircular end keyseat (Leven 1949; Okubo et al. 1968).

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Fillet radius.

In order to be conservative, in general, it is assumed a fatigue stress concentration factor of 2 on fillet radii.

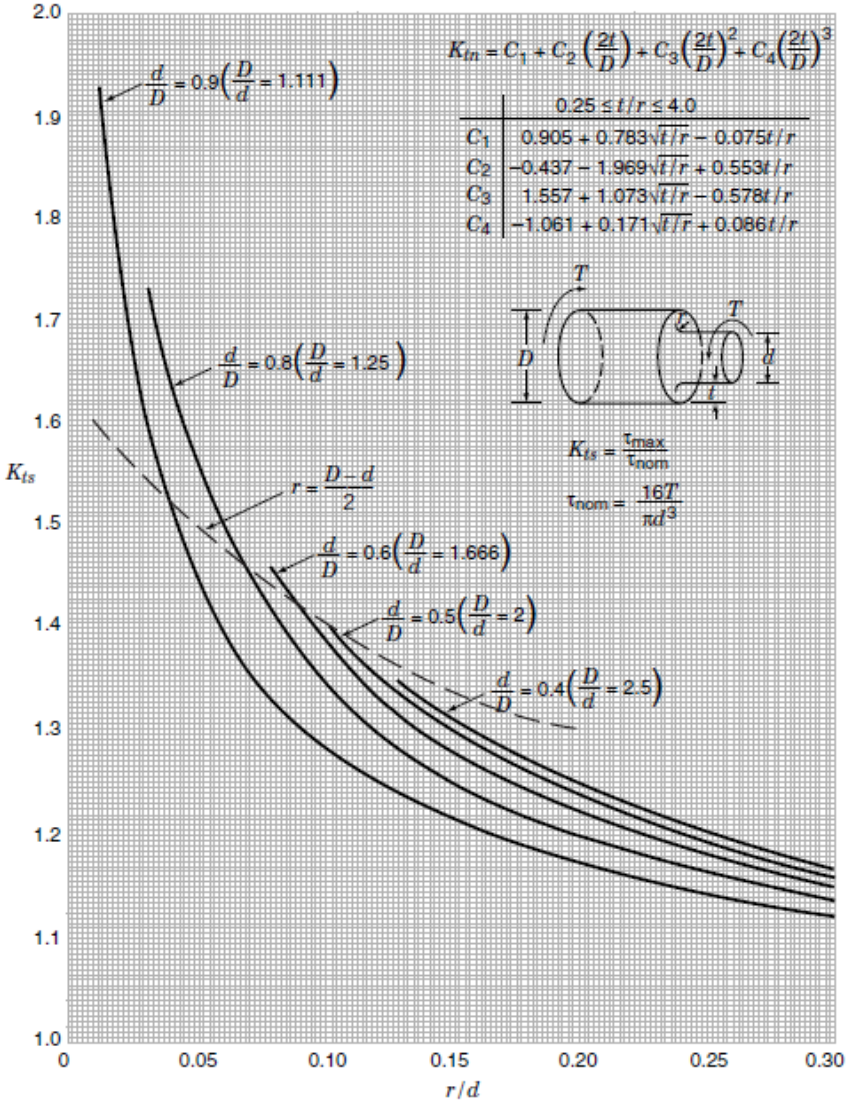

Chart 3.12 Stress concentration factors K_{ts} for torsion of a shaft with a shoulder fillet (data from Matthews and Hooke 1971).

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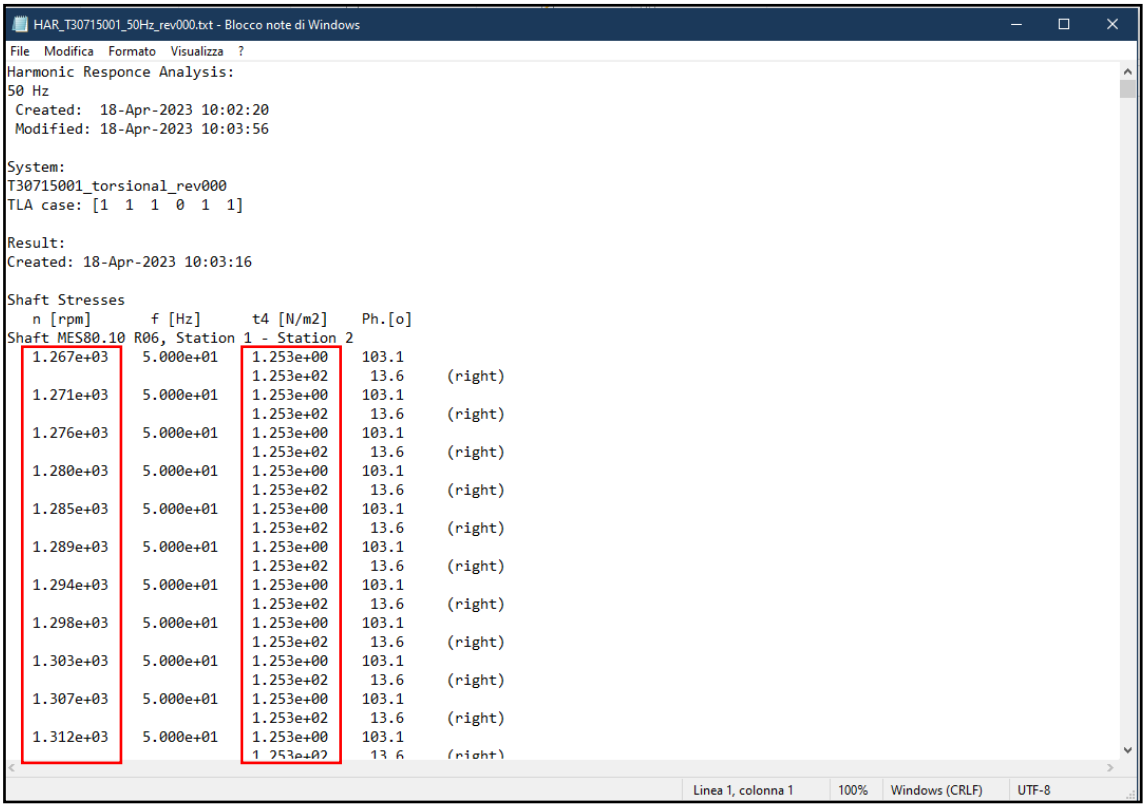
Alternate torsional stresses generated from the steady-stated response analysis can be extracted and stored for each excitation and for each shaft in a file.txt, as shown in Figure 33. The file will contain the shear stresses for each shaft element for each speed in the range considered (Figure 32). These torsional stresses will be summed for each excitation at the different rotating speed and the maximum of every station will be used to build the Goodman diagram. This process can be implemented with a script or a worksheet in Excel (*Mod.408 – Torsional analysis shaft fatigue verification*).



```
HAR_T30715001_50Hz_rev000.txt - Blocco note di Windows
File  Modifica  Formato  Visualizza  ?
Harmonic Responce Analysis:
50 Hz
 Created:  18-Apr-2023 10:02:20
 Modified: 18-Apr-2023 10:03:56

System:
T30715001_torsional_rev000
TLA case: [1  1  1  0  1  1]

Result:
Created: 18-Apr-2023 10:03:16

Shaft Stresses
   n [rpm]       f [Hz]      t4 [N/m2]    Ph.[o]
Shaft MES80.10 R06, Station 1 - Station 2
  1.267e+03    5.000e+01    1.253e+00    103.1
                            1.253e+02     13.6     (right)
  1.271e+03    5.000e+01    1.253e+00    103.1
                            1.253e+02     13.6     (right)
  1.276e+03    5.000e+01    1.253e+00    103.1
                            1.253e+02     13.6     (right)
  1.280e+03    5.000e+01    1.253e+00    103.1
                            1.253e+02     13.6     (right)
  1.285e+03    5.000e+01    1.253e+00    103.1
                            1.253e+02     13.6     (right)
  1.289e+03    5.000e+01    1.253e+00    103.1
                            1.253e+02     13.6     (right)
  1.294e+03    5.000e+01    1.253e+00    103.1
                            1.253e+02     13.6     (right)
  1.298e+03    5.000e+01    1.253e+00    103.1
                            1.253e+02     13.6     (right)
  1.303e+03    5.000e+01    1.253e+00    103.1
                            1.253e+02     13.6     (right)
  1.307e+03    5.000e+01    1.253e+00    103.1
                            1.253e+02     13.6     (right)
  1.312e+03    5.000e+01    1.253e+00    103.1
                            1.253e+02     13.6     (right)
Linea 1, colonna 1   100%   Windows (CRLF)   UTF-8
```

Figure 32. Torsional stress vs speed.

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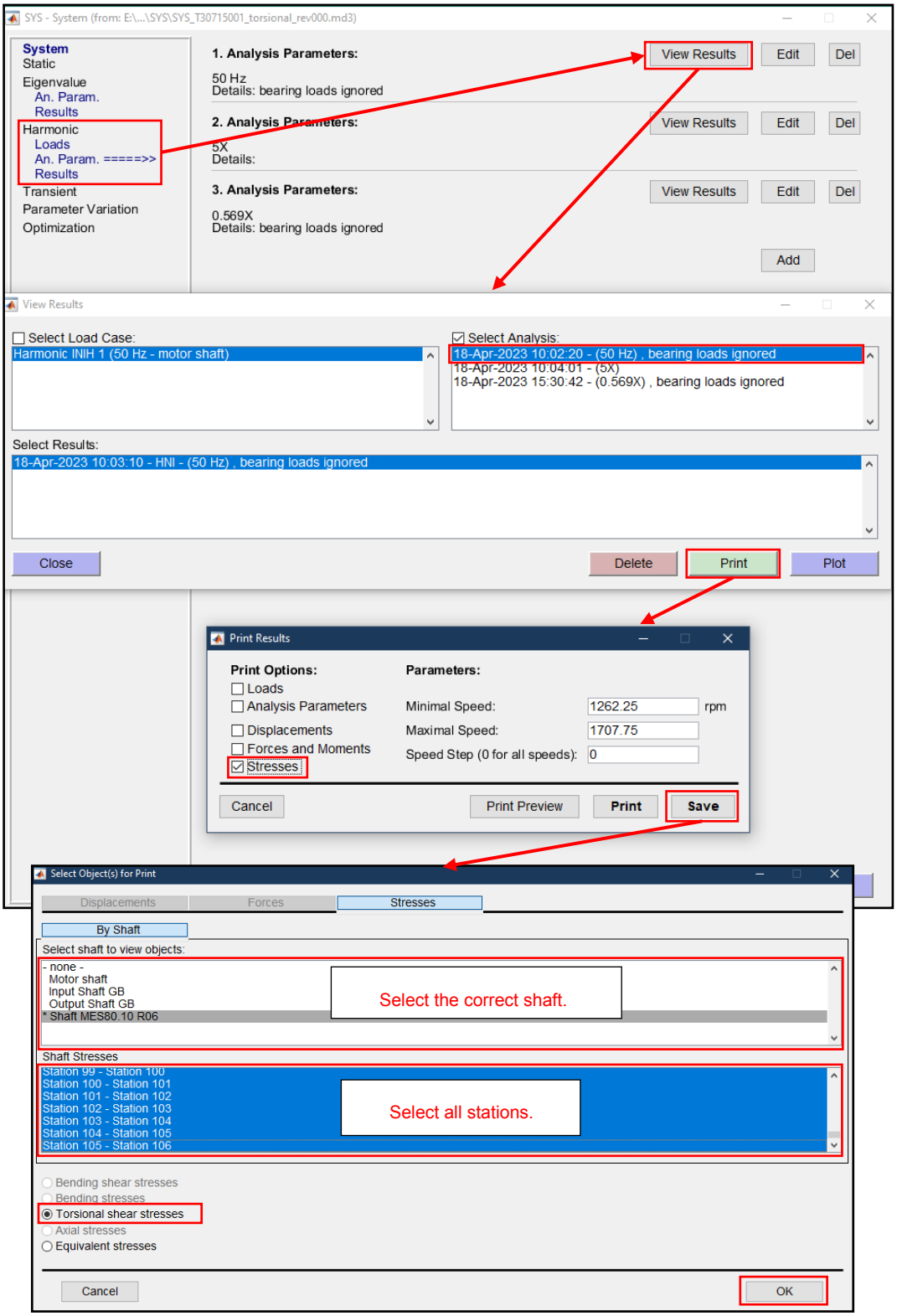

Figure 33. GUI to extract alternate stress.

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For the **static stress** due to the average torque a copy of System needs to be created because a static analysis will be performed and a main change in the model is mandatory. This important change is the creation of one constraint that should be applied to one extreme of the train where the torque is generated (motor core middle station) or the torque is absorbed (last impeller station to transmit torque, suggested). The constraint is modelled as a "**General spring**" with infinite torsional stiffness (Figure 34).

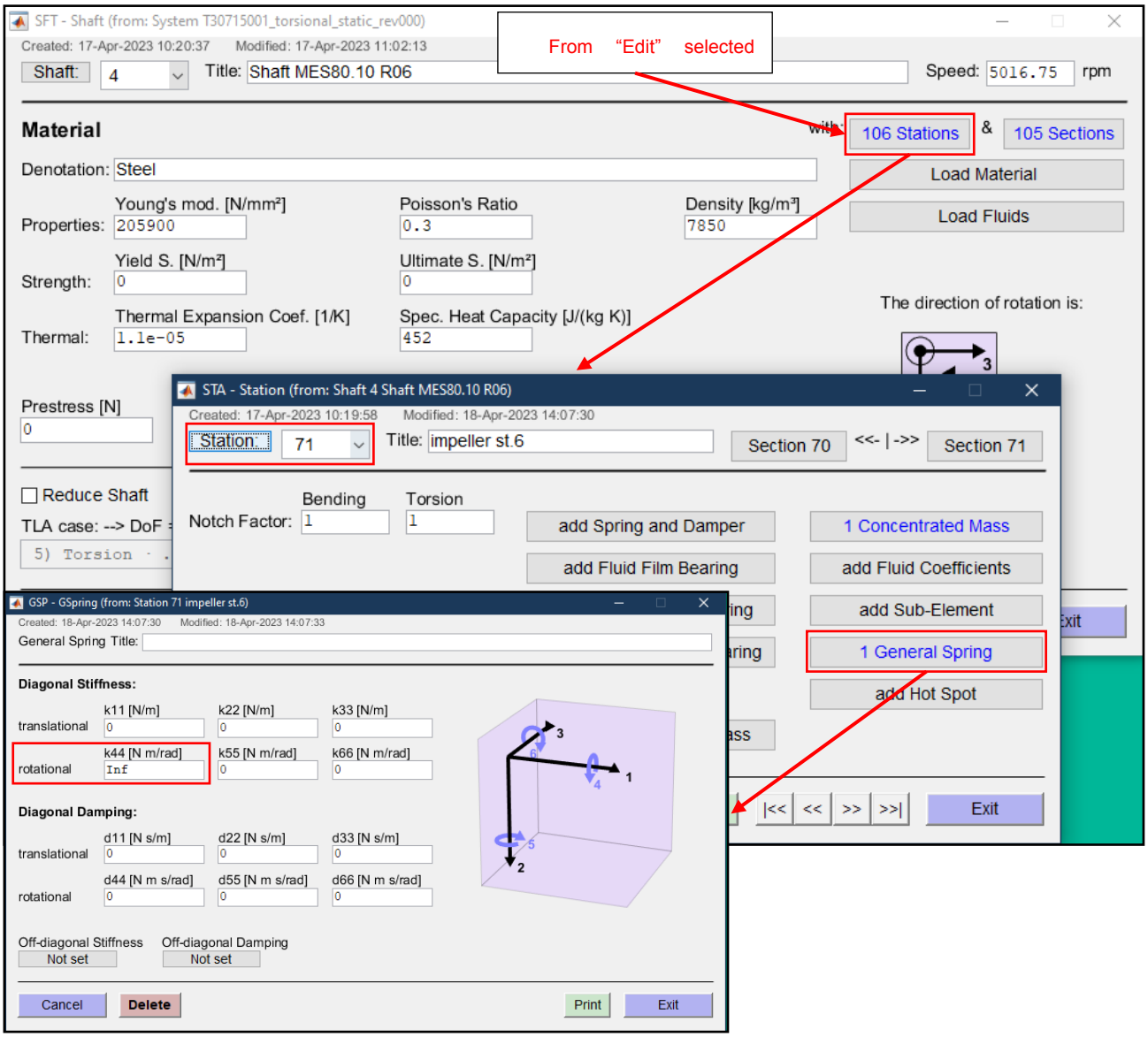

Figure 34. GUI constraint for static torsional analysis.

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The next step is the creation of the load cases for the analysis. In the middle of the motor core it will be applied the average torque (load case 1), while, on every pump impeller, the average torque divided by the number of stages (load case 2). To take into account the presence of both load cases a “static load case combination” will be created (Figure 35).

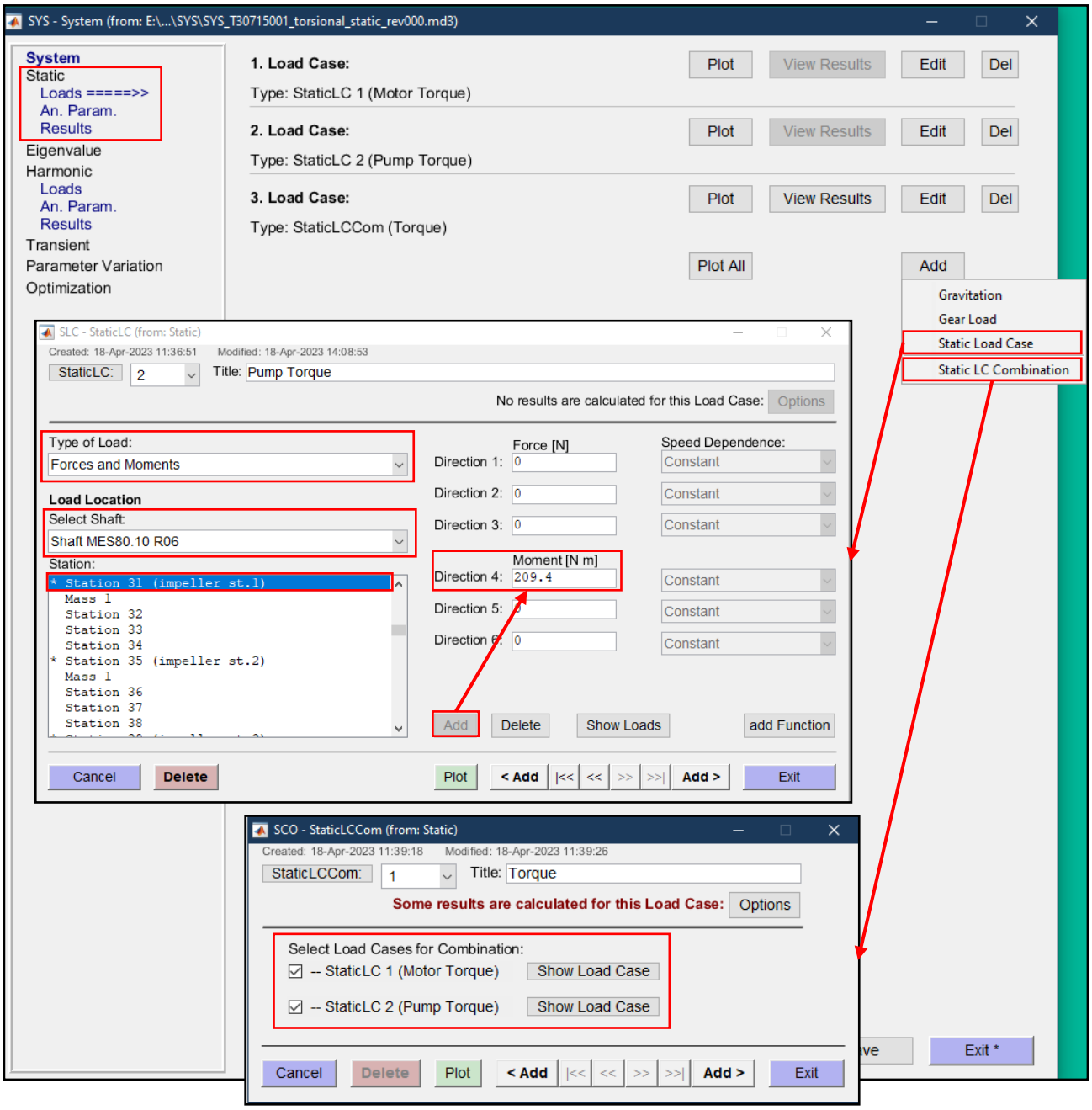

Figure 35. GUI load cases for static analysis.

NOTE. Since the torque on motor core is generated and the torque on impellers is absorbed, pump torque should have opposite sign to motor torque. For example, if motor torque is defined positive, pump torque should be negative. An exception is when the pump sense of rotation is the

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opposite of the motor one (i.e., with gear box). In this case the sign should be adjusted to the sense of rotation.

Figure 36 shows the correct steps for performing the static analysis.

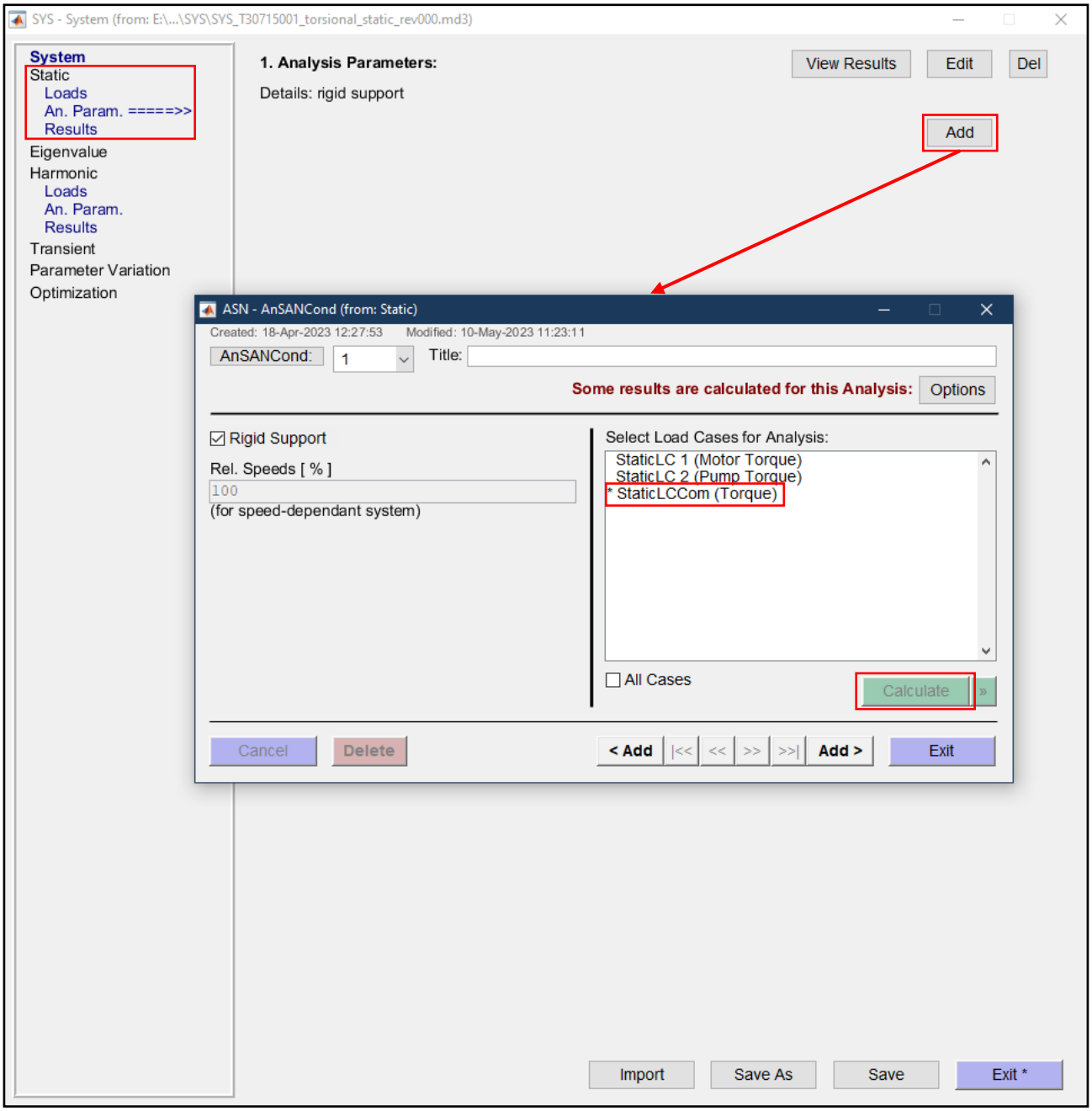

Figure 36. GUI static Analysis.

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Like the alternate stresses, results of static analysis can be extracted and stored in a file.txt, as show in Figure 37. The file will contain the equivalent shear stresses for each element in the shaft selected (Figure 38). These values will represent the mean torsional stress in the Goodman diagram.

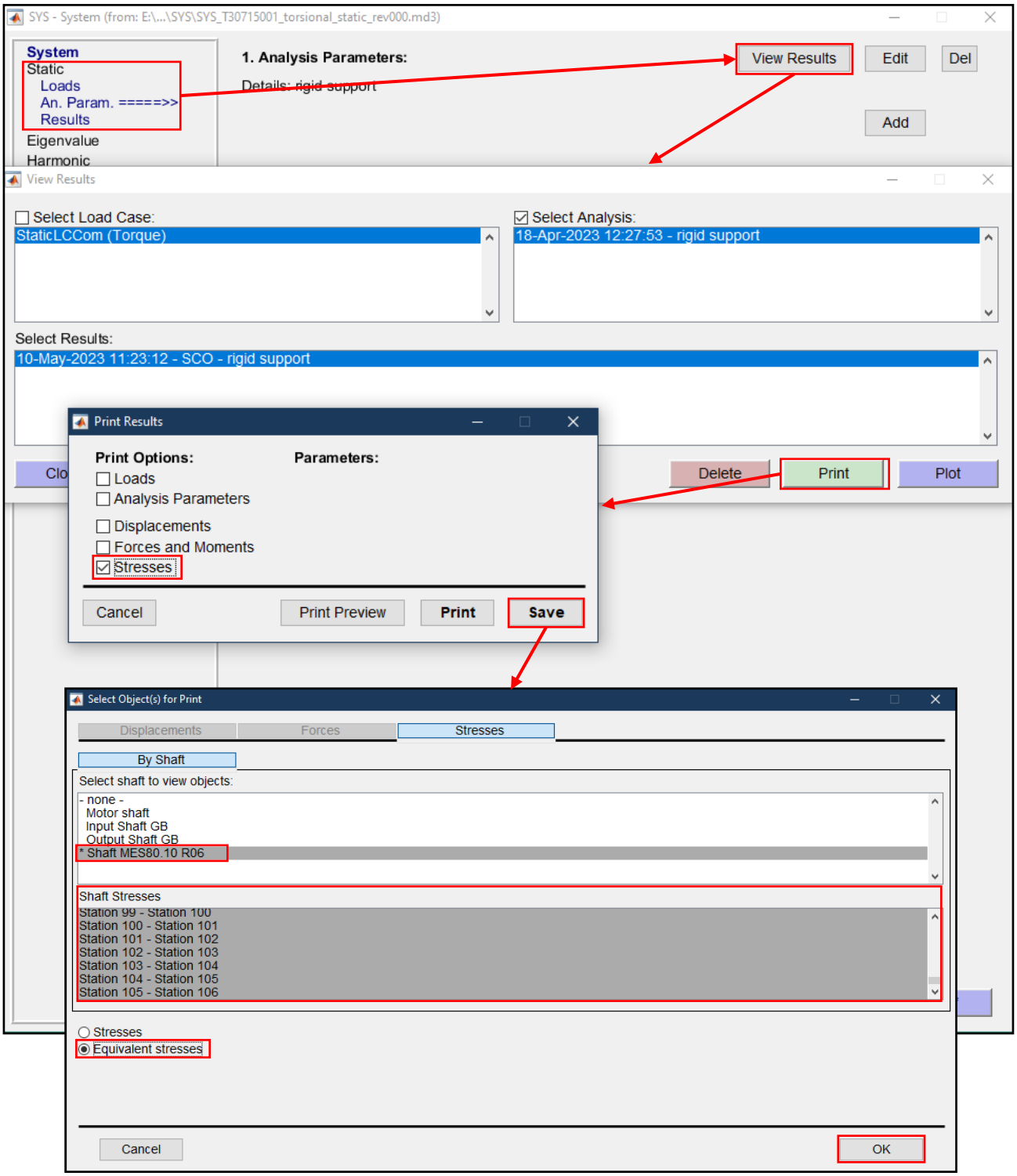

Figure 37. GUI to extract mean stresses.

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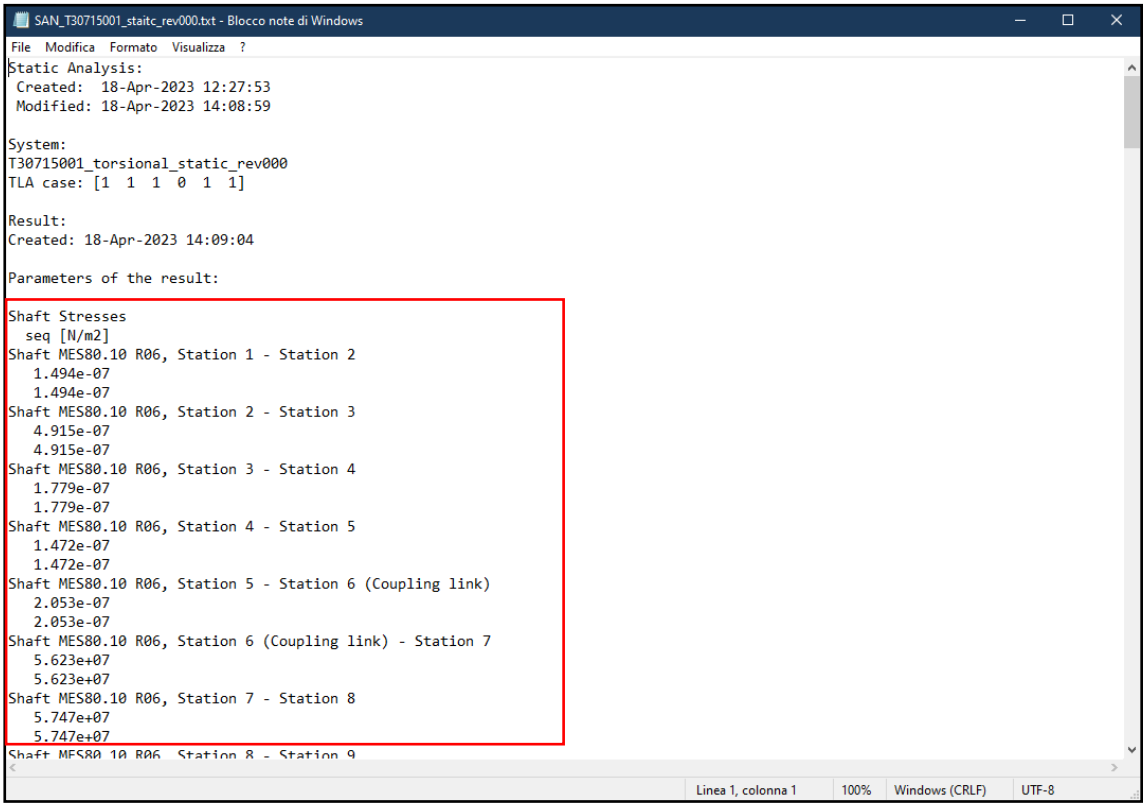
NOTE. In this case data were exported as equivalent stresses because the script used for fatigue analysis wants in input these values. However, in the Goodman diagram there are the nominal stress of mean and alternate shear. The equivalent stresses should be divided by $\sqrt{3}$.

$$\sigma_{eq.} = \sqrt{3}\tau_{nom.}$$


SAN_T30715001_staitc_rev000.txt - Blocco note di Windows

File Modifica Formato Visualizza ?

```
Static Analysis:
 Created:  18-Apr-2023 12:27:53
 Modified: 18-Apr-2023 14:08:59

System:
T30715001_torsional_static_rev000
TLA case: [1  1  1  0  1  1]

Result:
Created: 18-Apr-2023 14:09:04

Parameters of the result:

Shaft Stresses
  seq [N/m2]
Shaft MES80.10 R06, Station 1 - Station 2
   1.494e-07
   1.494e-07
Shaft MES80.10 R06, Station 2 - Station 3
   4.915e-07
   4.915e-07
Shaft MES80.10 R06, Station 3 - Station 4
   1.779e-07
   1.779e-07
Shaft MES80.10 R06, Station 4 - Station 5
   1.472e-07
   1.472e-07
Shaft MES80.10 R06, Station 5 - Station 6 (Coupling link)
   2.053e-07
   2.053e-07
Shaft MES80.10 R06, Station 6 (Coupling link) - Station 7
   5.623e+07
   5.623e+07
Shaft MES80.10 R06, Station 7 - Station 8
   5.747e+07
   5.747e+07
Shaft MES80.10 R06, Station 8 - Station 9
```

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Figure 38. Mean torsional stresses.

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The nominal alternate and mean stresses obtained for each station should be multiplied by the appropriate fatigue stress concentration factor (K_f) and then plotted in the Goodman diagram, as shown in Figure 39.

The acceptance criteria for shafts fatigue analysis is that all points (shaft stations) should be inside the allowable region.

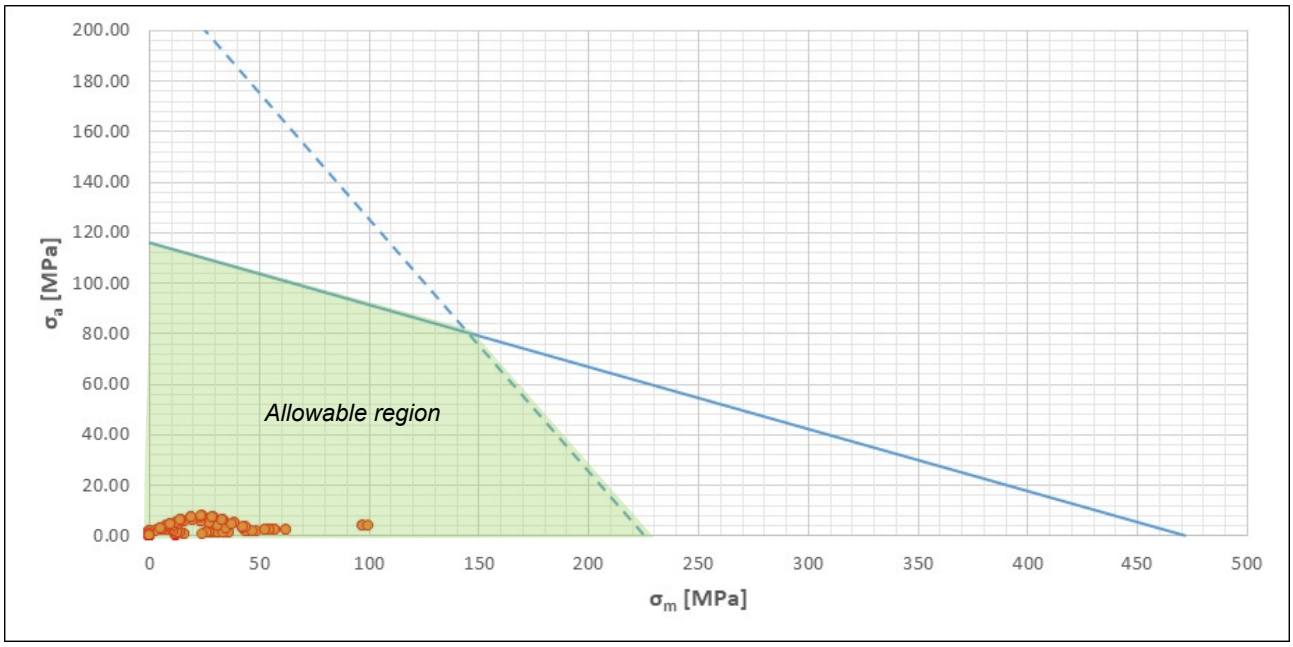

Figure 39. Goodman diagram.

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2.8. Coupling, gears, … verification.

Another check to make is the peak torque occurring in components such as gears and couplings. The peak torque (T_{peak}) is equal to the sum of the mean transmitted torque and the alternate torque. Most gears and couplings are rated for the maximum continuous torque (T_{rating}) that they may carry. It is the torque value at which the coupling can safely run continuously. Thus, it is recommended to keep the peak torque below the continuous rating for these components.

$$T_{peak} = T_{mean} + T_{alternate} < T_{rating}$$

The **mean torque (T_{mean})** is equal to:

$$T_{mean} = \frac{P}{\left(2\pi\frac{N}{60}\right)}$$

Where P is the rated power in [W] and N is the rotating speed in [rpm].

The **alternate torque ($T_{alternate}$)** is equal to the maximum, in the operating speed range, of the sum of all torsional critical speed responses.

$$T_{alternate} = \max_{[speed\ range]}\left(\sum_i T_{alternate,\,i}\right)$$

Where i represents all the torsional critical speeds in the operating speed range.

If the fatigue torque ($T_{fatigue}$) is available on the coupling drawing, the alternate torque must be compared to the fatigue torque and it must not be exceeded. The fatigue torque is the value that the coupling can handle without experiencing fatigue failure.

$$T_{alternate} < T_{fatigue}$$

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Figure 40 shows the envelope of the alternate torque. At each speed it is equal to the sum of the responses to all critical speeds.

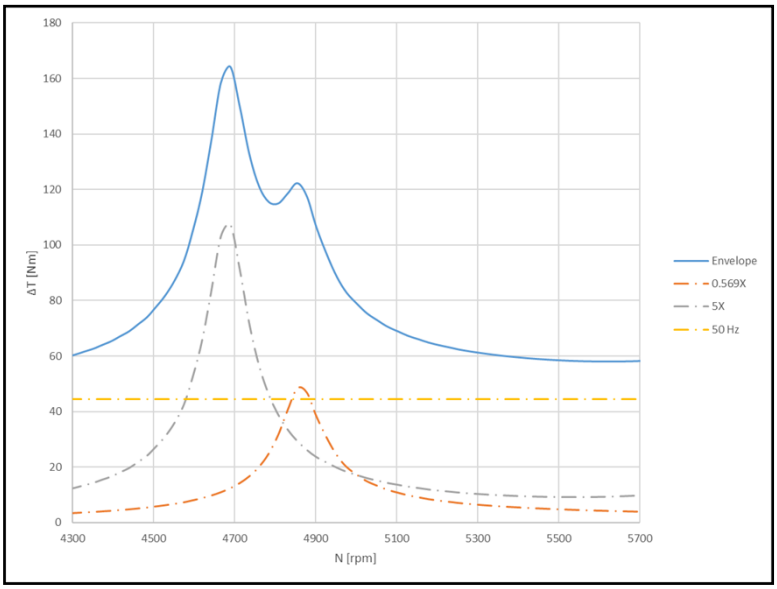

Figure 40. Example of alternate torque.

It is possible to extract and store the alternate torque for each excitation in a file.txt as shown in Figure 41. Then, these values (Figure 42) can be used to create the graph in Figure 40. The process can be implemented with a script or a worksheet in Excel (*Mod.409 – Torsional analysis coupling verification*).

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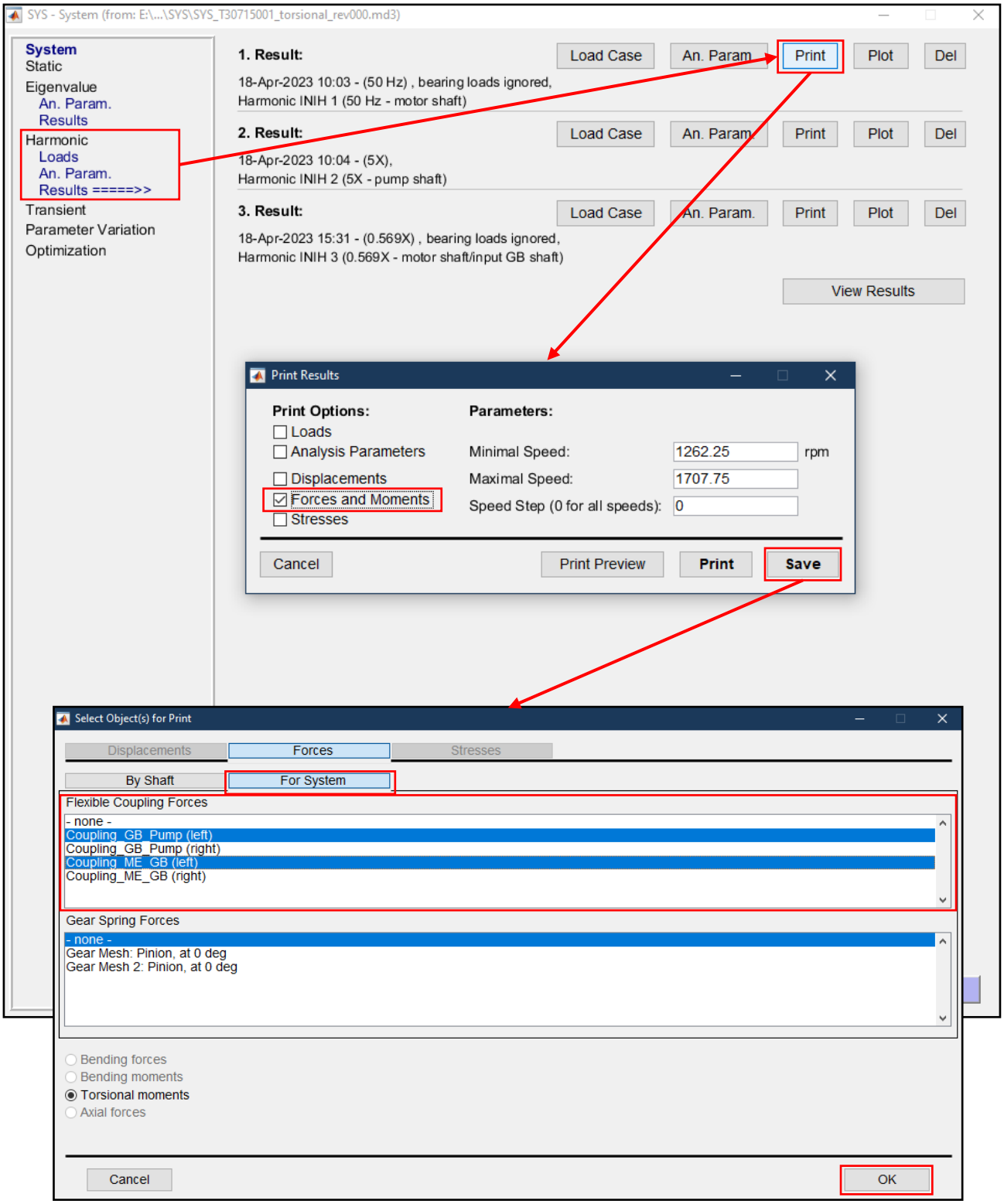

Figure 41. GUI to extract alternate torque

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HAR_T30715001_1X_coupling_rev001.txt - Blocco note di Windows

File Modifica Formato Visualizza ?

```
Harmonic Responce Analysis:
1X - pump shaft
 Created:  18-Apr-2023 10:04:01
 Modified: 10-May-2023 10:49:19

System:
T30715001_torsional_rev001
TLA case: [1  1  1  0  1  1]

Result:
Created: 10-May-2023 10:29:45

Flexible Coupling Forces
   n [rpm]        f [Hz]        M4 [Nm]      Ph.[o]
Coupling_GB_Pump (left)
   1.269e+03    7.143e+01    1.156e+01    173.2
   1.275e+03    7.178e+01    1.214e+01    173.2
   1.281e+03    7.214e+01    1.275e+01    173.3
   1.288e+03    7.250e+01    1.338e+01    173.3
   1.294e+03    7.287e+01    1.405e+01    173.3
   1.301e+03    7.323e+01    1.474e+01    173.3
   1.307e+03    7.360e+01    1.548e+01    173.2
   1.314e+03    7.396e+01    1.626e+01    173.2
   1.320e+03    7.433e+01    1.708e+01    173.1
   1.327e+03    7.471e+01    1.796e+01    173.0
   1.333e+03    7.508e+01    1.890e+01    172.9
   1.340e+03    7.545e+01    1.990e+01    172.8
   1.347e+03    7.583e+01    2.098e+01    172.7
   1.354e+03    7.621e+01    2.214e+01    172.5
   1.360e+03    7.659e+01    2.340e+01    172.4
   1.367e+03    7.697e+01    2.477e+01    172.2
   1.374e+03    7.736e+01    2.627e+01    171.9
   1.381e+03    7.775e+01    2.792e+01    171.7
   1.388e+03    7.813e+01    2.975e+01    171.4
   1.395e+03    7.853e+01    3.178e+01    171.0
   1.402e+03    7.892e+01    3.405e+01    170.6
   1.409e+03    7.931e+01    3.662e+01    170.2
```

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Figure 42. Coupling alternate torque vs. speed

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Annex D. Report structure.

Report should include:

1. INTRODUCTION.

- a brief description of the machine in study (type, train components and rotating speed);
- the purpose of the study: to provide acceptable torsional performance of the train in study over the operating speed range according to the last API Standard;
- the criteria to satisfy (API 610 §6.9.2.6 and §6.9.2.7).

1.1. REFERENCES. Documentation and computer program used.

2. NUMERICAL MODEL. It should provide the assumption made to model system and a schematic diagram of the model depicting nodes and elements.

2.1. EXCITATION SOURCES considered in the analysis.

3. NUMERICAL ANALYSIS. It should include a brief description of the method used to solve torsional calculations and the equation describing the dynamic behaviour of the system,

3.1. UNDAMPED NATURAL FREQUENCY ANALYSIS. A Campbell diagram should show all excitations sources, including strong multiples of the operating speed, and torsional natural frequencies. Torsional critical speeds, interference points between excitations and natural frequencies, should be highlighted.

A table should summarize the separation margin respect to the rated speed of each critical speed and those with a value less than 10% should be reported.

Figures of the torsional mode shapes should be included.

(If no critical speed has been reported, there is no need to perform a steady stated damped response analysis and following points).

3.2. STEADY STATE RESPONSE ANALYSIS. It should report the dynamic behaviour of the system subjected to the excitations identified. Displacement or stress results of the analysis should be included.

3.3. FATIGUE ANALYSIS OF PUMP SHAFT through a Goodman diagram. The Goodman diagram should report the alternate and mean stress of all stations in the system and these values should be placed in the allowable region to be accepted.

The paragraph should include all the parameters necessary for building the Goodman diagram:

- material properties (yield and strength stress);
- fatigue strength factors;

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- stress concentration factors.

3.4. COUPLING VERIFY. It should show that the peak torque (sum of the steady state torque and the dynamic torque) capability of coupling components, gearing and interference fits of components is lower than the nominal torque capability of the component.

A chart of vibratory torque vs speed helps to visualize the maximum alternate torque.

4. CONCLUSIONS. It should summarize the different results of all analysis done and state if the system meets the requirements.

5. ANNEX. It is a compendium of the numeric model.

5.1. MATERIAL PROPERTIES. A table that includes density, elastic modulus and Poisson's ratio of all materials used in the analysis.

5.2. SHAFT ELEMENTS. Tables that define the numerical model geometry (lengths, outer diameters, inner diameters of each element) of the shafts in the system.

5.3. INERTIA TABLE that defines the position and the inertia properties of the lumped masses.

5.4. ROTOR EQUIVALENT RIGID BODY PROPERTIES. A table that shows the main properties of each rotor (total length, mass, moment of polar inertia, speed ratio).

5.5. CONNECTIONS. Tables that define the different connections (coupling, gearmesh) used in the model and their main characteristics. For example, the starting and ending stations, the torsional stiffness and damping coefficient of a coupling.

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